Real Estate Management

Release 8.0

jankstar

Aug 25, 2026

CONTENTS

A Tryton ERP module for real estate management. Packaged as `jankstar_real_estate`, targeting Tryton 8.0.0 and Python 3.9–3.13.

Covers property management, lease and sales contracts, tenant/owner party roles, operating cost settlement, CO2 cost allocation under the German CO2KostAufG, and a German "Kontenrahmen der Wohnungswirtschaft" (WoWi) accounting chart.

CHAPTER

ONE

SETUP

CHAPTER

TWO

CONFIGURATION

This page describes every master-data record that must exist before the `real_estate` module can be used productively. Records marked **[auto]** are created automatically when the module is installed via `trytond-admin -u real_estate`. All other records must be set up manually or confirmed after installation.

2.1 SKR04 Chart of Accounts [auto]

The module ships a complete German SKR04 chart of accounts under `skr04/`. It is loaded automatically at installation and provides:

- Account types and accounts (e.g. 4120 Mieterträge, 1600 Forderungen)
- Tax groups, tax templates (7 % / 19 % MwSt.)
- Tax code templates and tax rule templates

After installation, open *Accounting* → *Configuration* → *Account Templates* and apply the chart to your company (button *Create Chart of Account*) if it has not been applied yet.

2.1.1 Rental Book Journal [auto]

A journal named **Rental Book** (code RE, type *Revenue*) is created automatically. It is used as the default posting journal for all contract types. If you prefer a different journal, you can change it on each contract type individually.

2.2 Account Configuration

Open *Accounting* → *Configuration* → *Configuration* and fill in the real-estate-specific defaults (added by this module via `account_configuration.py`, model `account.configuration.real_estate`):

Vacancy Cost Account (`re_account_allocation_by_owner`)
: Account used for direct GL postings of vacancy settlement results (cost shares with no contract, i.e. the owner's share of unoccupied periods) — both the debit and credit side of the posting.

Operating Cost Settlement Journal (`re_journal_billing`)
: Journal used for the vacancy GL postings above.

Operating Cost Billing Payment Term (`re_payment_term_billing`)
: Default payment term for operating cost settlement invoices, used whenever the contract itself has no payment term set.

`Vacancy Cost Account` and `Operating Cost Settlement Journal` are required before vacancy results can be billed (see `BillingUnit.billing_wizard` / `billing` in `billing_unit.py`); tenant invoices are unaffected and use the contract's own accounts/journal.

2.3 Use Classes [auto]

Six use classes are loaded automatically:

Seq.	Name	Basement No.	Parking No.
10	Apartment	yes	—
20	Office	yes	—
30	Retail	yes	—
40	Warehouse	yes	—
50	Parking	—	yes
60	Garage	—	yes

Additional classes can be created under *Real Estate* → *Configuration* → *Use Classes* without changing any code. The `has_basement_nr` / `has_parking_nr` flags control which extra fields appear on the rental-object form.

2.4 Measurement Types [auto]

The following measurement types are loaded automatically:

Seq.	Name	Unit	Applies to	Group
05	Usable Space	m²	object	yes (root)
10	Living Space	m²	object	child of *Usable Space*
15	Commercial Space	m²	object	child of *Usable Space*
20	Number of rooms	unit	object	—
30	Gross floor area	m²	building	—
40	Land area	m²	land	—
50	Number of items	unit	equipment	—
60	Property value	EUR	property	—

Usable Space is a group type: when it is referenced in a term type or settlement unit, the system automatically includes measurements of all child types (*Living Space*, *Commercial Space*) in the calculation.

Additional types can be created under *Real Estate* → *Configuration* → *Measurement Types*. All children of a group must share the same unit; circular parent references are rejected.

2.5 Object Party Roles [auto]

Three roles are loaded automatically:

Seq.	Name	Applies to object types
10	Caretaker (default)	property
20	Administrator	property
30	Owner	property, object

Additional roles can be created under *Real Estate* → *Configuration* → *Object Party Roles*.

2.6 Contract Types [auto]

Six contract types are loaded automatically:

Seq.	Name	Prefix	Direction	Occupancy	Type of use
10	Rental agreement	1	Debit (out)	yes	residential
20	Parking Space Lease Agreement	2	Debit (out)	yes	residential
30	Debit Costs Agreement	3	Debit (out)	—	all
40	Credit Costs Agreement	4	Credit (in)	—	all
50	Commercial lease agreement	5	Debit (out)	yes	commercial
80	Condominium fee agreement	8	Debit (out)	yes	property

For each contract type you should check and set, if not already done:

- **Default account** — the revenue/expense account for invoice lines
- **Default taxes** — e.g. 19 % USt. for commercial contracts
- **Payment term** — optional default payment term

2.7 Contract Term Types [auto]

Seven term types are loaded automatically:

Seq.	Name	Rhythm	Type of use / m_type
1000	Apartment rent (monthly)	1 × monthly	residential / Living Space
1100	Parking space rent (monthly)	1 × monthly	residential, commercial, internal / —
2000	AP for Operating costs (monthly)	1 × monthly	all / Living Space
3000	AP for Heating costs (monthly)	1 × monthly	all / Living Space
8000	Commercial rent (monthly)	1 × monthly	commercial / Commercial Space
9000	Condominium fees (monthly)	1 × monthly	property / —

For each term type you can optionally set:

- **Default account** — pre-filled on the contract term when this type is selected
- **m_type** — measurement type used to derive the default quantity (the value is looked up from the referenced contract item's objects)

2.8 Cost Category Groups [auto]

Five groups are loaded automatically, following the BetrKV paragraph structure:

- I · Grundbesitzabgaben & Versicherungen (seq. 100)
- II · Ver- und Entsorgung (seq. 200)
- III · Reinigung, Pflege & Sicherheit (seq. 300)
- IV · Sonstige Betriebskosten (seq. 400)
- V · Keine Umlage (seq. 900)

2.9 Cost Types (§ 2 BetrKV) [auto]

The following cost types are loaded automatically, covering the standard positions of § 2 BetrKV:

Seq.	Name (§ 2 BetrKV)	Group
100	Grundsteuer (Nr. 1)	I
110	Gebäudeversicherung (Nr. 13)	I
200	Wasserversorgung, Abwasser (Nr. 2 + 3)	II
300	Heizung – Brennstoff (Nr. 4)	II
310	Heizung – Wartung (Nr. 4)	II
320	Warmwasser (Nr. 5)	II
330	Verbundene Heizungs- und Warmwasserversorgung (Nr. 6)	II
400	Aufzug (Nr. 7)	II
500	Straßenreinigung (Nr. 8)	III
510	Müllabfuhr (Nr. 8)	III
520	Hausreinigung (Nr. 9)	III
600	Gartenpflege (Nr. 10)	III
610	Hausstrom (Nr. 11)	III
620	Schornsteinfeger (Nr. 12)	III
700	Hausmeister (Nr. 14)	IV

Additional cost types (e.g. antenna, broadband, communal laundry) can be added freely under *Real Estate* *Configuration* *Cost Types*.

2.10 Taxes

The SKR04 templates include tax records for 7 % and 19 % VAT. After applying the chart of accounts, verify that the tax record *USt. 19 % Umsatzsteuer voller Satz Waren Inland* exists under *Accounting* *Taxes*. This tax is used by `test_contracts.py` for commercial leases and should be assigned to the *Commercial lease agreement* contract type as a default tax.

CHAPTER

THREE

USAGE

This page describes the end-to-end workflow for the `real_estate` module, from setting up a property to running the annual operating cost settlement. All steps assume the configuration described in *Configuration* has been completed.

3.1 Step 1 — Create a Property

Navigate to *Real Estate* → *Properties*.

1. Click **New** and set **Type** to *Property*.
2. Fill in **Name**, **Company**, **Start date**, and optionally assign an **Address** (select from *Real Estate* → *Master Data* → *Addresses*).
3. Set **Type of use** (e.g. *Residential*) — this filters which contract types are available later.
4. Optionally set **Billing as** and **Collective billing** to control how operating cost billing is aggregated across buildings.
5. Click **Approved** to transition the property from *Draft* to *Approved* (button on the form toolbar). Further transitions are *Approved* → *Locked* → *Deactivated* (and back from *Locked* to *Approved*, or *Approved* to *Draft*).

Under the property you can then create the sub-objects of the tree:

Type	Parent	Typical use
Building	Property	Multi-unit residential or commercial building
Land	Property	Plot without building (e.g. parking lot)
Object	Building or Land	Rental unit: apartment, retail space, parking space
Equipment	Building, Land, Object, or Equipment	Meters (water, heat), technical installations

For each **Object** set the **Use class** (Apartment, Retail, Parking, …) and optionally enter the **Object number** (shown on invoices and reports).

For each **Equipment** of e_type *Meters* set:

- **Meter unit** — the UOM of the readings (e.g. m^3)
- **Meter factor** — conversion factor (usually 1.0)
- **No decimals** — tick if readings must be integer values
- **Meter ID** — identifier printed on the physical meter

Activate each object with the **Active** button after completing it.

3.2 Step 2 — Enter Measurements

Measurements record the physical size of an object for a given date. They are used as the allocation base for operating cost settlement (e.g. living area in m²) and as the default quantity on contract terms.

Navigate to the object's form and open the **Measurements** tab, or use *Real Estate* → *Master Data* → *Measurements*.

For each rental object enter at least:

- **Living Space** (m²) — used by *Apartment rent* and *Operating costs* term types as default quantity
- **Number of rooms** — optional, for reports

For commercial objects enter:

- **Commercial Space** (m²) — used by *Commercial rent* term types

For buildings:

- **Gross floor area** (m²) — used in billing unit reports

The **Valid from** date determines from when the measurement applies. Multiple measurements of the same type can coexist if the area changes over time; the system always uses the latest measurement valid on or before the reference date.

3.3 Step 3 — Assign Partners and Roles

Open the **Parties** tab on any property or object to assign `party.party` records with a role (e.g. Owner, Administrator).

For owners this is informational; for tenants the party is entered directly on the contract (Step 4).

3.4 Step 4 — Create Contracts

Navigate to *Real Estate* → *Contracts*.

1. Click **New** and set:
   - **Contract type** — e.g. *Rental agreement* (residential)
   - **Property** — the parent property
   - **Contractual partner** — the tenant (`party.party`)
   - **Invoice address** — the partner's delivery/invoice address
   - **Start date** — first day of the tenancy
   - **Currency** — filled automatically from the company
2. On the **Items** tab, click **New** to add a `ContractItem`:
   - **Valid from** — typically the contract start date
   - Add one or more rental objects in the **Objects** sub-table. The object selector is restricted to objects of type *object* that belong to the same property as the contract.
3. On the **Terms** tab, click **New** to add each charge (`ContractTerm`):
   - **Term type** — e.g. *Apartment rent (monthly)*
   - **Reference item** — the contract item this term refers to

- **Valid from** — start of the charge period
- **Rhythm** / **Rhythm type** — billing frequency (e.g. 1 × monthly)
- **Quantity** — auto-filled from the measurement type linked to the term type (can be overridden)
- **Unit price** — amount per unit per period
- **Taxes** — pre-filled from the contract type defaults

Typical term sequence for a residential contract:

Seq.	Term type	Note
10	Apartment rent (monthly)	Net rent
20	AP for Operating costs (monthly)	Cold costs advance
30	AP for Heating costs (monthly)	Heating advance

4. Click **Running** to activate the contract. The state changes from *Draft* *Running*.

3.4.1 Terminating a contract

Click **Terminate** on a running contract. The wizard asks for:

- **Terminated by** — tenant or landlord
- **Receipt of termination notice** — date the notice was received
- **Termination reason** — free text
- **Termination date** — last day of the tenancy (calculated from the notice period or entered manually)

The contract moves to *Terminated*. Occupancy records are updated automatically; the object becomes available for a new contract.

3.4.2 Cancelling a contract

The **Cancel** button is available on *Draft* contracts. If cash flow entries in state *done* (already invoiced) exist, a warning is shown: existing postings will **not** be reversed automatically. Draft cash flow entries are deleted automatically on cancellation.

3.5 Step 5 — Generate Invoices (CreateContractMoves)

Navigate to *Real Estate* *Create Contract Moves* or click the **Create Moves** button on a contract.

The wizard offers three actions:

`Create`
: Generate invoices up to a given cut-off date. For each contract term with pending future cash flow entries up to the date, one invoice line is created and the cash flow entry transitions to *done*.

`Re-calc`
: Rebuild the cash flow projection without creating invoices. Useful after changing a term's rhythm, unit price, or validity dates.

`Re-calc and Create`
: Rebuild first, then create invoices in one step.

You can filter by **Property** and/or **Contract** to limit the run to a subset of contracts.

After the wizard completes, the generated invoices appear under *Accounting* → *Invoices* and can be posted and sent from there.

The **Cash Flow** tabs on the contract form show:

- **Draft** — planned entries not yet invoiced
- **Pending** — posted (open) invoices
- **Paid** — settled invoices

3.6 Step 6 — Meter Readings

Navigate to *Real Estate* → *Master Data* → *Meter Readings*, or open the **Meters** tab on an equipment object.

Reading types:

`initial`
: First reading when the meter is commissioned (or a new tenant moves in). Must be the first reading for a given meter.

`reading`
: Periodic reading. The meter ID must match the previous reading. For counter meters, the **Consumption** is computed automatically (current value minus previous value).

`estimate`
: Created by the **Estimate Consumption** wizard on the equipment form. The wizard interpolates between surrounding readings or extrapolates from the last two readings within one year.

`final`
: Last reading when a meter is decommissioned or replaced.

The **Meter ID** groups readings that belong to the same physical meter. When a meter is replaced, a new initial reading with a different Meter ID starts a new series.

3.7 Step 7 — Operating Cost Settlement (Billing Unit)

The annual settlement is managed through *Billing Units*. Navigate to *Real Estate* → *Operation Costs* → *Billing Units*.

3.7.1 7.1 Create a Billing Unit

1. Click **New** and set:
   - **Property** — the property to settle
   - **Start date** — first day of the settlement period (e.g. 01.01.2025)
   - **End date** — last day (e.g. 31.12.2025); leave blank for open-ended
   - **Calculation method** — *Rental apartment* (§§ 1–2 BetrKV) or *WEG billing* (condominium law)
   - **Description** — e.g. “Kalte Betriebskosten 2025”
   - **Term types of use** — which contract term types count as advance payments (e.g. *AP for Operating costs*)
2. Click **Approved** to activate the billing unit.

3.7.2 7.2 Add Settlement Units

Under the **Settlement Units** tab, add one row per cost type:

- **Cost type** — e.g. *Grundsteuer* (seq. 100)
- **Allocation rule**:

  `allocation_by_measurement`
  : Allocated proportionally to a measurement value (e.g. living area). Set **Measurement type** — e.g. *Usable Space* (group, expands to Living Space + Commercial Space automatically).

  `allocation_by_consumption`
  : Allocated by meter reading consumption (e.g. water m³). Set **Meter unit** and **Meter regex** to filter the relevant meters.

  `allocation_per_rental_unit`
  : Allocated proportionally to occupancy duration.

  `allocation_from_external_billing`
  : No allocation calculation; amounts are entered manually from an external bill (e.g. heat cost allocator result).

  `no_allocation`
  : Entire cost stays unallocated (e.g. maintenance reserve).

- **Object regex** — regular expression to filter which rental objects are included (e.g. `Wohnung|Einzelhandel`).
- **Vacancy** — how to handle unoccupied periods: *by_owner* (costs go to the owner) or *unallocated*.

3.7.3 7.3 Run Selection

Click **Selection** to identify which contracts and objects fall within the settlement period. The system creates `CostShare` records for every object × settlement unit combination.

Selection can be re-run from the *Value Share* state to revise the scope; existing settlement results are deleted after confirmation.

3.7.4 7.4 Compute Value Shares

Click **Compute Value Shares** to calculate the allocation factor for each cost share:

- Measurement-based: `value_share = measurement × time_share / time_total`
- Consumption-based: `value_share = consumption`
- Per rental unit: `value_share = time_share / time_total`

Cost shares that cannot be computed (e.g. missing meter reading) are set to state *error* with an explanatory message. Fix the underlying data and click **Compute Value Shares** again — error shares are automatically reset before the recalculation.

3.7.5 7.5 Compute Settlement Result

Click **Compute Settlement Result** to aggregate cost shares per contract and produce one `SettlementResult` per contract:

- **Planned costs** — budgeted amount (from cost share)
- **Actual costs** — real costs entered or allocated

- **Advanced payment** — sum of *posted* and *paid* invoice lines for the term types of use during the settlement period
- **Refund / receivable** = actual costs − advanced payment (positive = refund to tenant, negative = additional charge)

If any cash flow entries in the settlement period are still in state *draft* or *validated* (not yet posted), the affected cost share is marked *error* with a `[draft]` prefix. Post or delete those invoice drafts and re-run.

3.7.6 7.6 Ready for Billing

Click **Ready for Billing** to transition from *Value Share* to *Ready for Billing*. This validates, for every settlement unit and settlement result of the billing unit, that:

- no settlement unit is in an error state;
- every settlement unit (except *No allocation*) has completed its value share calculation;
- at least one approved settlement result exists and none has actual costs of zero;
- every settlement result with a non-zero advance payment has both a term and a contract assigned (a missing term usually means several term types contributed and the billing unit's term-type filter needs narrowing);
- all advance-payment invoices are posted;
- unless **External billing** is enabled, the sum of settlement result actual costs matches the sum of cost share actual costs.

Any failed check raises a validation error listing the offending records — fix the underlying data and click **Ready for Billing** again. Without this step the **Billing** button (7.7 below) does not appear. This step can also be triggered for all billing units sharing the same settlement period via the **Ready for Billing** button on the property form.

3.7.7 7.7 Billing

Click **Billing** to create invoices and credit notes from the settlement results. For each contract with a settlement result the system creates:

- A credit note line reversing the advance payments booked during the year
- An invoice line for the actual costs allocated to the contract
- Optionally a journal entry for the owner allocation (WEG billing)

All postings of one billing run share a `billing_run_id` (`YYYYMMDD-HHMMSS-U<userid>`) for traceability.

If the property has **Collective billing** enabled, the *Billing* button is not shown on individual billing units; instead use the **Billing Property** button on the property form to trigger all billing units at once.

3.8 Typical Annual Workflow Summary

#	Action	Where
1	Create / update properties and objects	*Real Estate* *Properties*
2	Enter measurements (area, rooms)	Object form Measurements tab
3	Enter meter readings (initial, periodic)	*Real Estate* *Master Data* *Meter Readings*
4	Create / update contracts and terms	*Real Estate* *Contracts*
5	Generate invoices monthly (CreateContract-Moves)	*Real Estate* *Create Contract Moves*
6	Post invoices in accounting	*Accounting* *Invoices*
7	Create billing unit for the settlement year	*Real Estate* *Operation Costs* *Billing Units*
8	Add settlement units (cost types + allocation rules)	Billing unit form Settlement Units tab
9	Run Selection Compute Value Shares	Billing unit form (buttons)
10	Enter actual costs on settlement units	Settlement unit form
11	Compute Settlement Result	Billing unit form (button)
12	Click Ready for Billing	Billing unit form button / property form
13	Run Billing post invoices / credit notes	Billing unit form button / property form

CHAPTER

FOUR

DESIGN

This page describes the key architectural decisions, data model relationships, and invariants of the `real_estate` module. It is aimed at developers who want to extend or maintain the module.

4.1 Object Hierarchy

All real estate assets are represented by a single self-referential model `real_estate.base_object` using Tryton's `tree()` mixin. A single table stores all object types; the `type` field (`property`, `building`, `land`, `object`, `equipment`) distinguishes them.

Allowed parent-child combinations:

Object type	Valid parent types
property	— (root, no parent)
building	property, building
land	property
object	property, building, land
equipment	building, land, object, equipment

The `property` back-reference (stored Many2One, auto-computed from the path) points to the root of the tree. It is used extensively as a filter in contract items, billing units, and occupancy records.

The `path` field (Tryton `tree()` mixin) stores the full ancestor chain as a slash-separated string of IDs. SQL index on `path` enables fast subtree queries without recursive CTEs.

4.1.1 Single-table design rationale

Using one table rather than separate tables per type simplifies cross-type queries (e.g. "all objects under a property"), reduces join depth, and keeps the view/form layer uniform. Type-specific fields (meter fields, rental-object fields) are simply hidden by PYSON `states.invisible` conditions on the form view; they are NULL in the database for inapplicable types.

4.2 Measurement Group Hierarchy

`real_estate.measurement.type` supports an optional parent-group relationship: a type with `is_group = True` acts as a container for child types. Groups can be nested arbitrarily (multi-level hierarchies, e.g. *Usable Space* □ *Living Space* + *Commercial Space*).

Two helper methods encapsulate hierarchy traversal:

`get_hierarchy_ids()` (instance method)
Returns `[self.id]` + all descendant IDs recursively, including intermediate group nodes. Used in `ContractTerm.get_term_measurements` to find all measurements relevant to a group m_type.

`get_effective_ids(cls, m_type)` (classmethod)
Returns only **leaf** (non-group) IDs, recursively expanded. Used in `ContractTerm._sum_measurements` and `SettlementUnit._compute_value_shares` so that allocation queries hit only concrete measurement records, not group containers.

Constraints enforced by `validate_fields`:

- A type may not reference itself or any of its descendants as `parent` (cycle detection via ancestor walk).
- All children must share the same UOM as their parent group.
- Only leaf types (`is_group = False`) may be assigned to individual `real_estate.measurement` records on objects.

4.3 Contract Architecture

A contract is assembled from three layers:

```
Contract (real_estate.contract)
  └── ContractItem  (real_estate.contract.item)          – "what is rented"
        └── ContractItemObject                            – rental objects
  └── ContractTerm  (real_estate.contract.term)          – "what is charged"
        └── ContractTermTax                              – taxes
        └── ContractTermCashFlow                         – cash flow entries
```

ContractItem links one or more `BaseObject` records of type `object` to the contract for a given validity period. The object selector is domain-restricted to the same property as the contract (`property = Eval('property')`), preventing cross-property assignments.

ContractTerm defines a recurring charge. A term always references a `ContractItem` (`reference_item`) so that measurement-based quantity derivation and operating cost allocation can trace back to the physical object.

ContractTermCashFlow is the projection / ledger entry:

- `state = 'draft'` — planned, no invoice yet
- `state = 'done'` — linked to an `account.invoice.line`

Cash flow entries carry `create_moves_run_id` (`YYYYMMDD-HHMMSS-U<uid>`) so every posting can be traced to the exact wizard run that created it.

4.4 Contract Workflow

```
Draft ──► Running ──► Terminated
  │           │
  └───────────┴──► Cancelled
```

Transitions:

- **Draft** → **Running** (button *Running*): activates the contract; sets `running_by`; triggers occupancy refresh.

- **Running → Terminated** (wizard *Terminate Contract*): records `termination_date`, `receipt_of_termination_notice`, `terminated_by`; triggers occupancy refresh.
- **Terminated → Running**: re-activates a contract (e.g. after a termination was withdrawn).
- **Draft / Running → Cancelled** (button *Cancel*): shows `ContractCancelWarning` if any `done` cash flow entries exist; deletes `draft` cash flow entries; triggers occupancy refresh. Cancelled contracts are excluded from all occupancy calculations.

4.5 Occupancy Tracking

`real_estate.base_object.occupancy` is a derived, read-only table that records the rental state of each `object`-type asset over time.

It is **not** maintained manually. Instead it is recomputed automatically whenever:

- A `ContractItem` or `ContractItemObject` is created, modified, or deleted (`create` / `write` / `delete` hooks).
- A contract changes state (via `Contract.write()` which checks `_COMPUTE_VALUE_SHARES_FIELDS` — this set includes `'state'`).

The recomputation filters contracts by `state in ('draft', 'running', 'terminated')`; cancelled contracts are excluded automatically.

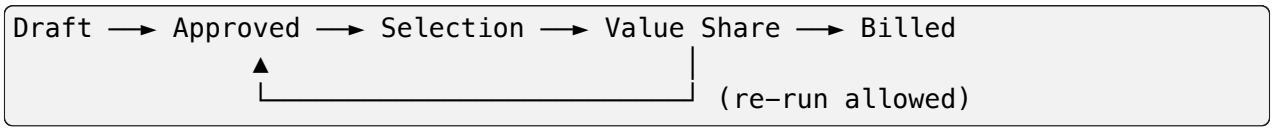
Occupancy states:

- **rented** — contract running, object occupied
- **under_negotiation** — contract draft, object provisionally reserved
- **vacant** — no active contract for the object

Overlap validation in `ContractItem._check_occupancy_overlap` rejects a second occupancy contract for the same object in the same period (hard error for `occupancy = True` contract types).

4.6 Operating Cost Settlement Pipeline

The settlement follows a strict state machine on `real_estate.billing_unit`:



Each stage builds on the previous:

1. **Approved** — billing unit is locked for editing; settlement units can be added.
2. **Selection** — `_run_selection()` identifies which objects and contracts fall inside the billing period. Creates one `CostShare` per (settlement_unit × object_or_contract) combination.
3. **Value Share** — `_compute_value_shares()` fills `CostShare.value_share` according to the allocation rule (measurement, consumption, per-unit, external, none). Errors are written to `CostShare.error_message`; the billing unit sub-state reflects whether errors exist.
4. **Billed** — `billing()` creates invoices and credit notes from `SettlementResult` records; writes `billing_run_id`.

Re-running *Selection* from *Value Share* deletes existing settlement results after a user confirmation warning (`SelectionWarning`).

Re-running *Compute Value Shares* automatically resets cost shares in state `error` to `selection` before recalculating, so corrected readings or measurements take effect without manual cleanup.

4.7 Allocation Rules

`SettlementUnit.allocation_rule` determines how costs are split:

`allocation_by_measurement`
Time-weighted share of a measurement value.

```
value_share = measurement_value × time_share / time_total
```

The `m_type` field can reference a **group** measurement type; the system expands it to all leaf descendants via `get_effective_ids()`. A unit-fallback applies for non-group types: if the exact type is not found on an object, the system falls back to any measurement with the same UOM.

`allocation_by_consumption`
Raw meter reading consumption (HeizkostenV).

```
value_share = end_reading − start_reading
```

Readings are looked up within a tolerance window (`reading_pre_days` / `reading_post_days` on the cost type). If no reading is found, the cost share transitions to `error`.

Vacancy fallback: when the immediately following contract cost share has no start reading, the system uses the previous vacancy's start reading as a substitute, allowing a single read-out at move-out to serve both periods.

`allocation_per_rental_unit`
Pure time share: `value_share = time_share / time_total`.

`allocation_from_external_billing`
No calculation. Amounts are entered manually on the cost share.

`no_allocation`
Entire cost remains unallocated.

4.8 Invoice Integration

The module extends three core accounting models:

`account.invoice`
Adds `contract` (Many2One to `real_estate.contract`) so every invoice is traceable to its origin contract.

`account.invoice.line`
Adds `term` (Many2One to `real_estate.contract.term`) and `settlement_unit` (Many2One to `real_estate.settlement_unit`). Term lines are created by `CreateContractMovesWizard`; settlement lines are created by `BillingUnit.billing()`.

`account.move.line`
Extended list view to expose `contract` and `term` context for journal entries.

`AccountContract` / `GeneralLedgerAccountContract` provide a contract-filtered view of the general ledger, accessible from the contract's *Account* tab.

4.9 Tryton Framework Patterns Used

`Workflow`
: Used by `BaseObject`, `Contract`, and `BillingUnit`. State transitions are declared in `__setup__` via `cls._transitions` and enforced by `@Workflow.transition` decorators on buttons.

`DeactivableMixin`
: Soft-delete for most master data and transactional records. The `active` flag is `True` by default; filtering in list views excludes inactive records automatically.

`tree()`
: Self-referential parent-child for `BaseObject` and `MeasurementType`. Provides `path` field and subtree search helpers.

`sequence_ordered()` / `re_sequence_ordered()`
: Integer `sequence` field with auto-increment helpers; used for consistent ordering in all One2Many tabs.

`TaxableMixin`
: Tax calculation on `Contract`; used to derive tax amounts for invoice line generation.

`Quantitative` (custom `fields.Numeric` subclass)
: Carries a `unit` reference and `quantitative=True` flag so the SAO client renders the associated UOM symbol next to the numeric value. Used for meter reading values and contract term quantities.

`Cache`
: `MeasurementType._get_default_type_cache` and `_get_window_domains_cache` are invalidated in `on_modification` whenever a measurement type record changes.

`UserWarning`
: `ContractCancelWarning` (cancel with existing postings), `SelectionWarning` (re-run selection), and `OccupancyOverlapWarning` follow the standard Tryton pattern: `Warning.format(key, records)` + `Warning.check(key)` + raise.

`@set_employee`
: Decorator used by workflow transitions to stamp `running_by`, `terminated_by`, and `cancelled_by` from the current user.

CHAPTER

FIVE

API REFERENCE

This page lists every model provided by the `real_estate` module together with its key fields, states, and notable methods. Extension fields added to Tryton core models are listed separately at the bottom.

Notation: **[R]** = readonly, **[F]** = Function field, **[R/O]** = required on save.

5.1 Property Management

5.1.1 `real_estate.use_class`

Configurable use class for rental objects (Apartment, Retail, Parking, …).

Field	Type	Description
`name`	Char [R/O]	Display name (translatable)
`sequence`	Integer [R/O]	Sort order; also used by demo scripts to find records
`has_basement_nr`	Boolean	Show *Basement Number* field on objects of this class
`has_parking_nr`	Boolean	Show *Parking Number* field on objects of this class

5.1.2 `real_estate.address`

Structured address with granular sub-fields.

Field	Type	Description
`street_name`	Char	Street name without number
`building_number`	Char	House / building number
`unit_number`	Char	Apartment / unit identifier
`floor_number`	Char	Floor
`room_number`	Char	Room identifier
`post_box`	Char	Post box number
`postal_code`	Char	Postal code
`city`	Char	City
`country`	Many2One	`country.country`
`street`	Char [F]	Composed `street_name + building_number` for display

5.1.3 real_estate.base_object

Central entity for every real estate asset.

Types: property · building · land · object · equipment

Workflow: Draft Approved Locked (+ Deactivated)

Field	Type	Description
name	Char [R/O]	Description / display name
type	Selection [R/O]	Object type; controls visible fields
type_of_use	Selection	residential / commercial / property / internal
use_class	Many2One	real_estate.use_class; visible for type *object* only
company	Many2One [R/O]	company.company
start_date	Date [R/O]	Date the asset was commissioned
end_date	Date	Date the asset was decommissioned
address	Many2One	real_estate.address
property	Many2One [R] [F]	Root property object (auto-derived from tree path)
parent	Many2One	Direct parent in the object tree
children	One2Many	Direct children
path	Char [R]	Full ancestor path (tree mixin)
state	Selection	draft / approved / locked / deactivated
object_number	Char [R]	Auto-generated object identifier
measurements	One2Many	real_estate.measurement records
parties	One2Many	real_estate.object_party (roles)
billing_units	One2Many	real_estate.billing_unit (property only)
occupancy	One2Many [R]	real_estate.base_object.occupancy (computed)
billing_as	Selection	single / collective — aggregation for cost billing
collective_billing	Boolean [F]	Derived from property's billing_as
year_of_constructio	Char	4-digit year (building only)
number_of_floors	Integer	Number of floors (building only)
floor	Integer	Floor number (object only)
basement_nr	Char	Basement storage number (object, use_class dependent)
parking_nr	Char	Parking number (object, use_class dependent)
e_type	Selection	Equipment sub-type: meters / other (equipment only)
meter_unit	Many2One	UOM for meter readings (equipment/meters only)
meter_is_counter	Boolean	True for cumulative counters (consumption = diff to prev. reading)
meter_factor	Float	Conversion factor applied to raw readings (default 1.0)
meter_no_decimals	Boolean	Readings must be integer values
meter_id	Char [F]	ID of the most recent reading
meter_last_value	Quantitative [F]	Last recorded reading value
meter_last_consumpt	Quantitative [F]	Consumption since the previous reading

5.1.4 real_estate.base_object.occupancy

Read-only occupancy ledger; recomputed automatically on contract changes.

Field	Type	Description
base_object	Many2One [R]	The object being tracked
property	Many2One [R]	Root property (denormalized)
start_date	Date [R]	First day of occupancy period
end_date	Date [R]	Last day (None = open-ended)
state	Selection [R]	rented / vacant / under_negotiation
contract	Many2One [R]	Source contract (None for vacancy periods)

5.1.5 real_estate.measurement.type

Named measurement type linked to a UOM. Supports group hierarchy.

Field	Type	Description
name	Char [R/O]	Display name (translatable)
unit	Many2One [R/O]	product.uom
types	MultiSelection	Object types this measurement applies to
is_group	Boolean	Marks this type as a group container
parent	Many2One	Parent group (is_group = True required on parent)
children	One2Many	Child types (visible when is_group = True)
default	Boolean	Use as default type for the object type
no_print	Boolean	Exclude from printed reports

Key methods:

get_hierarchy_ids()
: Returns self.id plus all descendant IDs recursively (including groups).

get_effective_ids(cls, m_type)
: Returns only leaf (non-group) IDs, recursively expanded from m_type.

5.1.6 real_estate.measurement

Associates a numeric value with a `BaseObject` for a given date.

Field	Type	Description
`base_object`	Many2One [R/O]	The object being measured
`m_type`	Many2One [R/O]	`real_estate.measurement.type` (leaf types only)
`valid_from`	Date [R/O]	Date from which this measurement applies
`value`	Float [R/O]	Measured value
`symbol`	Char [F]	UOM symbol (derived from m_type)
`no_print`	Boolean	Exclude from printed reports

5.1.7 real_estate.meter_reading

Meter reading linked to an equipment object of type *meters*.

Field	Type	Description
`base_object`	Many2One [R/O]	Equipment object (`e_type = 'meters'`)
`meter_id`	Char [R/O]	Physical meter identifier
`reading_date`	Date [R/O]	Date of reading
`m_type`	Selection [R/O]	`initial` / `reading` / `estimate` / `final`
`value`	Quantitative [R/O]	Raw meter value
`unit`	Many2One [F]	UOM (derived from equipment's `meter_unit`)
`consumption`	Quantitative [F]	Value minus previous reading (counter meters only)
`reading_user`	Many2One	User who recorded the reading
`comment`	Char	Free text

5.1.8 real_estate.object_party.role

Configurable role definition for object parties.

Fields: `name` (Char), `sequence` (Integer), `types` (MultiSelection of object types), `default` (Boolean).

5.1.9 real_estate.object_party

Links a `party.party` to a `BaseObject` with a typed role.

Fields: `base_object` (Many2One), `party` (Many2One), `role` (Many2One `object_party.role`), `valid_from` (Date), `valid_to` (Date).

5.2 Contract Management

5.2.1 real_estate.contract.type

Template controlling contract behaviour.

Field	Type	Description
name	Char [R/O]	Display name (translatable)
sequence	Integer [R/O]	Sort order
types_of_use	MultiSelection	Which type_of_use values this type applies to
invoice_type	Selection	out (debit/receivable) or in (credit/payable)
account	Many2One	Default party account for invoice lines
taxes	Many2Many	Default taxes applied to all terms
account_journal	Many2One	Default posting journal
prefix	Char [R/O]	Contract number prefix (e.g. 1)
start_number	Integer [R/O]	First contract number
step_item	Integer [R/O]	Sequence step for contract items
step_term	Integer [R/O]	Sequence step for contract terms
occupancy	Boolean	Enforce occupancy exclusivity for this contract type
mark	Char	Periodic posting mark (Debit / Credit)

5.2.2 real_estate.contract.term.type

Template for a recurring charge line.

Field	Type	Description
name	Char [R/O]	Display name (translatable)
sequence	Integer [R/O]	Sort order; also used by scripts to locate records
types_of_use	MultiSelection	Applicable type-of-use values
m_type	Many2One	Measurement type for default quantity derivation
default_quantity	Float	Fallback quantity when no measurement is found
account	Many2One	Default account for invoice lines
rhythm	Integer	Number of rhythm units per invoice
rhythm_type	Selection	daily / weekly / monthly / quarterly / annually / one_time
rhythm_start	Selection	Day of month (1–28) when the first invoice of the period is due

5.2.3 real_estate.contract

Main contract record.

Workflow: Draft Running Terminated / Cancelled

Field	Type	Description
c_type	Many2One [R/O]	real_estate.contract.type
property	Many2One [R/O]	Root property
contractual_partner	Many2One [R/O]	party.party (tenant / owner)
invoice_address	Many2One	party.address used on invoices
type_of_use	Selection	residential / commercial / property / internal
currency	Many2One [R/O]	Billing currency
start_date	Date [R/O]	Contract start
end_date	Date [F]	Effective end (max of item valid_to dates)
state	Selection	draft / running / terminated / cancelled
items	One2Many	real_estate.contract.item
terms	One2Many	real_estate.contract.term
logs	One2Many [R]	real_estate.contract.log
running_by	Many2One [R]	User who activated the contract
terminated_by	Many2One [R]	User who terminated the contract
cancelled_by	Many2One [R]	User who cancelled the contract
termination_date	Date	Last day of the tenancy
receipt_of_termina	Date	Date the termination notice was received
termination_reason	Text	Free-text reason

5.2.4 real_estate.contract.item

Assigns rental objects to a contract for a validity period.

Field	Type	Description
contract	Many2One [R/O]	Parent contract
label	Char	Optional label (overrides auto-name)
valid_from	Date [R/O]	Start of object assignment
valid_to	Date	End of object assignment (open if None)
objects	One2Many	real_estate.contract.item.object (rental objects)
property	Many2One [F]	Derived from contract.property

5.2.5 real_estate.contract.item.object

Junction between a `ContractItem` and a `BaseObject`.

Field	Type	Description
`item`	Many2One [R/O]	Parent `ContractItem`
`object`	Many2One [R/O]	`real_estate.base_object` (type *object*); domain-restricted to objects of the same property as the contract
`property`	Many2One [F]	Derived from `item.contract.property`

5.2.6 real_estate.contract.term

Recurring charge line on a contract.

Field	Type	Description
`contract`	Many2One [R/O]	Parent contract
`term_type`	Many2One [R/O]	`real_estate.contract.term.type`
`reference_item`	Many2One	`ContractItem` this charge relates to
`valid_from`	Date [R/O]	Start of charge period
`valid_to`	Date	End of charge period (open if None)
`rhythm`	Integer	Billing frequency count
`rhythm_type`	Selection	`daily` / `weekly` / `monthly` / `quarterly` / `annually` / `one_time`
`rhythm_start`	Selection	Day-of-month start (1–28)
`quantity`	Quantitative	Units per billing period
`unit`	Many2One [F]	UOM (from term_type.m_type.unit)
`unit_price`	Monetary	Price per unit per period
`taxes`	Many2Many	Applied taxes
`cash_flows`	One2Many	`real_estate.contract.term.cash_flow`

5.2.7 real_estate.contract.term.cash_flow

Single projected or realised cash flow entry.

Field	Type	Description
`term`	Many2One [R/O]	Parent `ContractTerm`
`state`	Selection	`draft` (planned) / `done` (invoiced)
`posting_date`	Date	Invoice / posting date
`document_date`	Date	Document date on the invoice
`due_date`	Date	Payment due date
`invoice_line`	Many2One [R]	`account.invoice.line` (set when state = done)
`create_moves_run_ic`	Char [R]	Run ID of the `CreateContractMoves` wizard invocation
`invoice_state`	Selection [F]	Derived from linked invoice: `draft` / `validated` / `posted` / `paid`

5.2.8 `real_estate.contract.log`

Append-only audit log entry on a contract.

Fields: `contract` (Many2One), `event` (Char), `description` (Text), `create_date` (DateTime [R]), `create_uid` (Many2One [R]).

5.3 Operating Cost Settlement

5.3.1 `real_estate.cost_category_group`

Groups cost types for reporting.

Fields: `name` (Char [R/O]), `sequence` (Integer [R/O]).

5.3.2 `real_estate.cost_type`

Individual operating cost item (e.g. *Grundsteuer*).

Field	Type	Description
`name`	Char [R/O]	Display name (translatable)
`sequence`	Integer [R/O]	Sort order; used by scripts to locate records
`category_group`	Many2One	`real_estate.cost_category_group`
`comment`	Text	BetrKV paragraph reference or free text
`no_print`	Boolean	Exclude from settlement printout
`reading_pre_days`	Integer	Days before target date within which a meter reading is accepted (default 7; `allocation_by_consumption` only)
`reading_post_days`	Integer	Days after target date within which a meter reading is accepted (default 7; `allocation_by_consumption` only)

5.3.3 `real_estate.billing_unit`

Annual (or period) operating cost settlement for one property.

Workflow: Draft Approved Selection Value Share Billed

Field	Type	Description
property	Many2One [R/O]	Root property
description	Char [R/O]	Free-text label (e.g. “Kalte Betriebskosten 2025”)
start_date	Date [R/O]	First day of settlement period
end_date	Date [F]	Last day (defaults to end of calendar year of start_date)
calculation_method	Selection	rental_apartment (§§ 1–2 BetrKV) / WEG_billing
billing_type	Selection	planned_billing / actual_billing
state	Selection	draft / approved / selection / value_share / billed
sub_state	Selection [F]	ok / error / warning (derived from cost share states)
term_types_of_use	MultiSelection	Term type sequences counted as advance payments
settlement_units	One2Many	real_estate.settlement_unit
settlment_results	One2Many	real_estate.settlement_result
moves	One2Many	real_estate.billing_unit.moves
billing_run_id	Char [R]	YYYYMMDD-HHMMSS-U<uid> stamped at end of billing()
sum_planned_costs	Monetary [F]	Sum of planned costs across all settlement units
sum_actual_costs	Monetary [F]	Sum of actual costs
sum_advanced_paymer	Monetary [F]	Sum of posted/paid advance payments
sum_refund_receivat	Monetary [F]	sum_actual_costs – sum_advanced_payment

5.3.4 real_estate.settlement_unit

One cost-type line within a billing unit.

Field	Type	Description
billing_unit	Many2One [R/O]	Parent billing unit
type	Many2One [R/O]	real_estate.cost_type
allocation_rule	Selection [R/O]	allocation_by_measurement / allocation_by_consumption / allocation_per_rental_unit / allocation_from_external_billing / no_allocation
vacancy	Selection	by_owner / unallocated
m_type	Many2One	Measurement type for measurement-based allocation
meter_unit	Many2One	UOM for consumption-based allocation
reg_ex_object	Char	Regex to filter rental objects (applied to base_object.name)
reg_ex_meter	Char	Regex to filter meter equipment (applied to base_object.name)
value_total	Float	Total allocation base (sum of all cost shares’ value_share)
cost_shares	One2Many	real_estate.cost_share

5.3.5 real_estate.cost_share

Intermediate allocation record: share of one cost type for one contract/object within a billing period.

States: `preparation` → `selection` → `estimated_value_share` → `value_share` · `error`

Field	Type	Description
`settlement_unit`	Many2One [R/O]	Parent settlement unit
`contract`	Many2One	Contract occupying the object (None = vacancy)
`base_object`	Many2One	The rental object
`start_date`	Date	Start of this share's occupancy period
`end_date`	Date	End of this share's occupancy period
`state`	Selection	See states above
`value_share`	Float	Allocation factor (measurement value × time fraction, or raw consumption, or time fraction)
`time_share`	Integer [F]	Days within the billing period
`allocation_rule`	Selection [F]	Mirrored from the parent settlement unit
`external_billing`	Boolean [F]	Mirrored from the parent settlement unit's billing unit
`planned_costs`	Monetary	Budgeted cost for this share. Readonly unless `external_billing` is set (otherwise computed, not entered manually)
`actual_costs`	Monetary	Real cost for this share. Readonly unless `external_billing` is set (otherwise computed, not entered manually)
`error_message`	Char [R]	Human-readable error (set when state = error)

5.3.6 real_estate.settlement_result

Final per-contract settlement for one billing unit.

States: `approved` → `billed`

Field	Type	Description
`billing_unit`	Many2One [R/O]	Parent billing unit
`contract`	Many2One	Tenant contract
`base_object`	Many2One	Rental object (denormalized)
`start_date`	Date	Settlement period start
`end_date`	Date	Settlement period end
`state`	Selection	`approved` / `billed`
`planned_costs`	Monetary	Sum of planned costs from cost shares
`actual_costs`	Monetary	Sum of actual costs from cost shares
`advanced_payment`	Monetary	Sum of posted/paid advance invoice lines
`refund_receivable`	Monetary [F]	actual_costs − advanced_payment (positive = refund)
`term`	Many2One	The single advance-payment term (if unambiguous)
`invoice`	Many2One [R]	Settlement invoice created by `billing()`

5.3.7 real_estate.billing_unit.moves

One record per posting created by `billing()`.

Field	Type	Description
billing_unit	Many2One [R/O]	Parent billing unit
settlement_result	Many2One	Source settlement result
contract	Many2One	Source contract
billing_run_id	Char [R]	Same value as parent BillingUnit.billing_run_id
moves_advanced_payn	Many2One	Credit-note line reversing advance payments
moves_actual_costs	Many2One	Invoice line for actual costs
moves_alloc_by_owne	Many2One	Journal move line for owner allocation (WEG only)

5.4 Extensions to Core Modules

party.party (party.py)
Adds sequence (Integer, sort key) and salutation (Char).

account.invoice (invoice.py)
Adds contract (Many2One → real_estate.contract).

account.invoice.line (invoice.py)
Adds real-estate assignment fields, gated by assignment_control (Selection: *All*, contract, operating_costs, settlement_result_contract, settlement_result_vacant): contract / term (originating contract / term), base_object, billing_unit / settlement_unit (for operating-cost lines), property [F], service_period_from / service_period_to, estg_35a (Selection, German §35a EStG tax-deduction category), and invoice_date / tax_amount / total_amount [F].

account.move.line (invoice.py, class AccountMoveLine)
Mirrors the same fields as account.invoice.line above (assignment_control, contract, term, base_object, billing_unit, settlement_unit, property [F]) directly on the journal entry line.

account.configuration (account_configuration.py)
Adds three real-estate defaults, all Many2One: re_account_allocation_by_owner (Vacancy Cost Account, used for direct GL postings of vacancy settlement results), re_journal_billing (journal for those postings), and re_payment_term_billing (default payment term for operating cost settlement invoices).

res.user (res.py)
Adds phone (Char) and mobile (Char).

5.5 Wizards

real_estate.contract.create_moves.wizard
Batch invoice generation. Start state start collects create_date, optional property and contract filters, and action (create / re_calc / re_calc_and_create).

real_estate.terminate_contract.wizard
Terminates a running contract. Collects terminated_by_type,

receipt_of_termination_notice, termination_reason, and termination_date (or computes it from a notice period in months).

real_estate.estimate_consumption.wizard
Two-step wizard on meter equipment objects. Step 1 collects per_date, meter_id, and reason; step 2 shows the simulated consumption and lets the user confirm or adjust the estimated_value before saving a new MeterReading.

CHAPTER

SIX

RELEASE NOTES

CHAPTER

SEVEN

MODULE DEPENDENCIES

ir, res, company, party, product, currency, country, account, account_invoice, account_deposit, account_payment_clearing, account_tax_non_deductible

CHAPTER

EIGHT

INSTALLATION

```
pip install -e .           # development install
pip install -e '.[test]'   # with test dependencies
```

The module entry point is declared in `setup.py`:

```
[trytond.modules]
real_estate = trytond.modules.real_estate
```

Link or install the package inside the Tryton modules directory so that `trytond-admin -u real_estate` can find it.

CHAPTER

NINE

CONFIGURATION

The following master data must be set up before the module can be used:

real_estate.object_party.role
: Partner roles in combination with the property type (`base_object.type`), e.g. *Tenant*, *Owner*, *Administrator*.

real_estate.measurement.type
: Named measurements linked to a unit of measure, e.g. *Living Area (m²)*, *Number of Rooms*, *Gross Floor Area (m²)*.

real_estate.contract.type
: Contract types defining invoice direction (in/out), default journal, tax defaults, contract number prefix, and whether occupancy exclusivity is enforced.

real_estate.contract.term.type
: Term type definitions with default rhythm (monthly / quarterly / …), default quantity source (measurement type), and default account.

real_estate.use_class
: Dynamic use-class catalogue replacing the former static selection field. Each record carries two boolean flags:

  has_basement_nr
  : Show/hide the *Basement Number* field on rental objects.

  has_parking_nr
  : Show/hide the *Parking Number* field on rental objects.

  Six default records are loaded at module installation: *Apartment* (has_basement_nr), *Office* (has_basement_nr), *Retail* (has_basement_nr), *Warehouse* (has_basement_nr), *Parking* (has_parking_nr), *Garage* (has_parking_nr). Additional classes can be created without changing code.

real_estate.cost_category_group / real_estate.cost_type
: Cost categories (e.g. *Heating*, *Water*) and individual cost types used to structure operating cost settlements. Default values for German BetrKV (§ 2) are loaded at module installation.

real_estate.re_accounting (re_accounting.py)
: Standalone, company-scoped real-estate accounting configuration, referenced one-directionally from `company.company.re_accounting` (`company.py`; optional — a company without a linked `re_accounting` record gets none of the automation below).

  re_account_allocation_by_owner
  : Vacancy cost account (debit and credit side of vacancy postings).

  re_journal_billing
  : Journal used for direct GL postings in operating cost settlements.

re_payment_term_billing
Default payment term for operating cost settlement invoices, used when the contract itself has none set.

create_moves_horizon_days
Number of days ahead of today the update_contract_cash_flow cron task (see below) recalculates cash flow for. Default 60.

co2_landlord_share_commercial
Default CO2 cost landlord share (%, 0–100) for commercial properties, which are not covered by the residential 10-tier distribution model — see *CO2 Cost Allocation (CO2KostAufG)* under *Operating Cost Settlement* below. Shown right before the *Cron Tasks* table on the form.

cron_tasks
One2Many to real_estate.cron_task — the company's scheduled task configuration, see below.

real_estate.cron_task (cron_task.py)
Per-re_accounting (i.e. per-company) row configuring one recurring operation: task (selection, currently update_contract_status / update_contract_cash_flow / book_contract_cash_flow / update_option_rate), valid_from, valid_until, interval_days, interval_months, schedule_day_of_month, horizon_months_ahead, last_run (updated by the dispatcher), active. Task names are deliberately rhythm-agnostic (no "daily"/"rolling"/… in the name) — how often each task actually runs is a per-row configuration choice made by the user, not implied by the task itself; see *Scheduling and parameters* below. None of interval_days/interval_months/ schedule_day_of_month is individually required, but validate() rejects a row that has none of the three set. A unique SQL constraint prevents duplicate rows for the same (re_accounting, task) pair. No rows are seeded automatically — a company gets no automatic behaviour until at least one row is added under the *Cron Tasks* tab of its real_estate.re_accounting record.

Dispatch. A single ir.cron entry ("Real Estate Daily Tasks", method real_estate.contract|cron_daily — registered via a small ir.cron.method selection extension in ir.py, since that field is otherwise a fixed core list) runs daily and calls Contract.cron_daily() (contract_core.py). It iterates all active cron_task rows and, for each one that is due (see *Scheduling* below), calls the matching Contract._cron_<task>(re_accounting, task) classmethod and updates last_run. A future recurring operation only needs a new _cron_<code> classmethod plus a new get_tasks() option — no new ir.cron entry.

Scheduling and parameters. Each cron_task row carries its own parameters, read by its handler from the task record passed in, and its own rhythm - e.g. update_contract_status daily (interval_days = 1), update_contract_cash_flow every six months (interval_months = 6), book_contract_cash_flow and update_option_rate monthly (schedule_day_of_month set, e.g. 15 and 1 respectively).

valid_from/valid_until (both optional) are hard lower/upper bounds checked before all three modes below - the task never runs while today < valid_from or today > valid_until. If the task has never run yet and today is already past valid_from, it becomes due immediately (using today, not valid_from, as the actual first last_run). E.g. setting valid_from = 15.03.2026 on a row activated on 20.03.2026 makes it run right away on 20.03.2026, not wait until the next scheduled date. valid_until just stops execution once passed - the row itself, and its last_run history, stay in place. validate() rejects a row where valid_until is before valid_from.

Three scheduling modes, checked by Contract._cron_task_is_due() in this priority order:

1. **Day-of-month-based**: if schedule_day_of_month (1-31) is set, it takes priority over both interval fields — the task runs at most once a month, on or after that calendar day (capped to the last day of shorter months), guarded by comparing last_run's year/month to today's so a

delayed run still catches up without re-running twice in the same month.

2. **Month-interval-based**: otherwise, if `interval_months` is set, it takes priority over `interval_days`. If `valid_from` is also set, `Contract._add_months()` computes the exact recurring date `valid_from + k * interval_months` months (same day-of-month as `valid_from`, clamped to shorter months), advancing `k` past `last_run` each time - so e.g. `valid_from = 15.03.2026` with `interval_months = 1` runs on 15.03., 15.04., 15.05., … and re-aligns to the correct slot even after a delayed run, instead of drifting from whatever date the delayed run actually happened on. Without `valid_from`, falls back to a looser check: due once at least that many calendar months (year/month difference, not exact days) have passed since `last_run` - a convenience for longer rhythms (e.g. 6 for half-yearly) without day-of-month precision.
3. **Day-interval-based** (default): otherwise, due once `today - last_run >= interval_days` (or `last_run` is unset).

`_cron_update_contract_status`
Auto-terminates `running` contracts of the company whose `end_date` has passed and which have no active termination yet (`termination_date` unset): sets `state = 'terminated'`, `termination_date = end_date`, `terminated_by_type = 'expired'`, and a `contract.log` entry — reusing the existing `terminated` state rather than adding a new one, so all state-dependent logic (occupancy, `create_moves`, item validation) keeps working unchanged. See `get_effective_end_date()` (`termination_date` if set and earlier than `end_date`, else `end_date`) which already governs occupancy for both `running` and `terminated` contracts regardless of whether this task has run yet.

`_cron_update_contract_cash_flow`
Cash-flow recalculation only (`action='re_calc'`): calls `Contract.call_create_moves()` for all running/terminated contracts of the company up to `today + create_moves_horizon_days`, so each term's `ContractTermCashFlow` stays generated ahead of time. It does **not** create invoices — booking is a separate, explicitly scheduled step, see `_cron_book_contract_cash_flow` below.

`_cron_book_contract_cash_flow`
Books (creates draft invoices for) all due terms, using `action='re_calc_and_create'` so it is self-sufficient regardless of whether `_cron_update_contract_cash_flow`'s horizon already covers the target date. The horizon is the end of the month that is `horizon_months_ahead` months after the run date — e.g. with `schedule_day_of_month = 15` and `horizon_months_ahead = 1`, a run on 15.07 books everything due up to 31.08. Invoices are created in `draft` state — this task does **not** post them; posting remains a manual step.

`_cron_update_option_rate`
Recomputes and, where the rate actually changed, books new option rates via `OptionRate.process_update()` (`option_rate.py` — the same logic as the manual `real_estate.option_rate_update.wizard`, see *Wizards* below) for every property of every company using this `re_accounting` configuration, as of today's date. Intended to run monthly with `schedule_day_of_month = 1`, which also determines the `effective_date` used by `process_update` (the 1st of the current month). Per-record results (created/updated/unchanged/ skipped counts, plus one detail line per processed object) go to the Python logger, not `contract.log`, since this task is not tied to individual contracts.

CHAPTER

TEN

ACCESS CONTROL

Four user groups are provided. Access is additive — a user may belong to multiple groups.

`group_real_estate_admin` — Real Estate Administration
: Full CRUD on all module models. Intended for system administrators and property managers with unrestricted access. The built-in `admin` user is automatically assigned to this group at installation.

`group_real_estate_object` — Real Estate Object
: Full CRUD on property and object data (`base_object`, `address`, `measurement`, `meter_reading`, `object_party`, `occupancy`). Read-only access to contract, billing, and configuration models. Intended for facility managers who maintain the property master data but do not manage contracts or run settlements.

  Exception: also has full CRUD on `real_estate.option_rate` (in addition to `group_real_estate_admin`), since option rates are maintained together with the property/object master data via the *Update Option Rates* wizard — see *Option Rate* below. The `group_real_estate_contract` and `group_real_estate_billing` groups remain read-only on it.

`group_real_estate_contract` — Real Estate Contract
: Full CRUD on contract data (`contract`, `contract.term`, `contract.item`, `contract.log`, `contract.term.cash_flow`, `contract.term.tax`, `account_contract`). Read-only access to property, billing, and configuration models. Intended for property managers / letting agents.

  Exception: also has full CRUD on `real_estate.contract.term.adjustment` (in addition to `group_real_estate_admin`) — see *Contract Term Adjustment* below. The `group_real_estate_object` and `group_real_estate_billing` groups remain read-only on it.

`group_real_estate_billing` — Real Estate Billing
: Full CRUD on all operating cost and settlement models (`billing_unit`, `billing_unit.log`, `billing_unit.moves`, `settlement_unit`, `settlement_result`, `cost_share`, `cost_category_group`, `cost_type`, `account.configuration.real_estate`). Read-only access to property and contract models. Intended for the operating cost accountant who runs the annual settlement but does not modify contracts or property master data.

Permission matrix (C = CRUD · R = read · – = no access):

Model	admin	object	contract	billing
`real_estate.address`	C	C	R	R
`real_estate.base_object`	C	C	R	R
`real_estate.base_object.occupancy`	C	C	C	R
`real_estate.option_rate`	C	C	R	R

continues on next page

Table 1 – continued from previous page

Model	admin	object	contract	billing
real_estate.meter_reading	C	C	R	R
real_estate.measurement.type	C	R	R	R
real_estate.use_class	C	R	R	R
real_estate.measurement	C	C	R	R
real_estate.object_party	C	C	R	R
real_estate.object_party.role	C	R	R	R
real_estate.contract	C	R	C	R
real_estate.contract.item	C	R	C	R
real_estate.contract.term	C	R	C	R
real_estate.contract.term.tax	C	R	C	R
real_estate.contract.log	C	R	C	R
real_estate.contract.account_contract	C	R	C	R
real_estate.contract.term.cash_flow	C	R	C	R
real_estate.contract.term.adjustment	C	R	C	R
real_estate.account_contract	C	R	R	R
real_estate.contract.type	C	R	R	R
real_estate.contract.term.type	C	R	R	R
real_estate.re_accounting	C	R	R	R
real_estate.cron_task	C	R	R	R
real_estate.co2_emission_share	C	R	R	R
real_estate.contract.type.tax	C	—	R	R
real_estate.cost_category_group	C	R	R	C
real_estate.cost_type	C	R	R	C
real_estate.co2_kostaufg	C	R	R	C
real_estate.co2_kostaufg.consumption	C	R	R	C
real_estate.billing_unit	C	R	R	C
real_estate.billing_unit.log	C	R	R	C
real_estate.billing_unit.moves	C	R	R	C
real_estate.settlement_unit	C	R	R	C
real_estate.cost_share	C	C	C	C
real_estate.settlement_result	C	C	C	C
account.configuration.real_estate	C	R	R	R

CHAPTER

ELEVEN

DATA MODEL

11.1 Property Management

`real_estate.address` (`address.py`)
: Structured address with granular fields (`street_name`, `building_number`, `unit_number`, `floor_number`, `room_number`, `post_box`) plus an unstructured fallback text field. Supports both formats and is linked to `base_object` records.

`real_estate.base_object` (`base_object.py`)
: Central entity for every real estate asset.

Types: `property` · `building` · `object` · `land` · `equipment`

Workflow: Draft Active Closed

Key features:

- Parent–child tree (`tree()` mixin), e.g. Property Building Apartment
- One2Many relations to `Address`, `ObjectParty`, `Measurement`, and `BillingUnit`
- History tracking via `BaseObjectOccupancy` (tenant occupancy periods)
- Meter readings (`MeterReading`) linked to equipment objects
- `billing_as` / `collective_billing` flags to control how operating cost billing is aggregated at property level
- `next_billing_start_date` (function field, property only): the earliest `start_date` among all non-`billed` billing units of the property. Drives the *ready for billing* checks and the green line-color highlight in the billing unit list view.
- Buttons `compute_value_shares`, `compute_settlement_result_property`, and `ready_for_billing_property` delegate bulk settlement actions to all billing units of the property that share `next_billing_start_date`. `billing_property` opens the `real_estate.billing_unit.wizard` instead of billing directly.
- `call_billing` / `do_billing` (classmethods) run the actual billing for one or more properties, optionally in the background queue (`execute_in_queue`). `do_billing` requires every billing unit that would be billed to already be in state `ready_for_billing`; with `collective_billing` all billing units sharing the same start date must be included together, otherwise a `ValidationError` is raised.

Rental object fields (visible only for `type = 'object'`):

`type_of_use`
: Economic use: `residential`, `commercial`, `property` (owner-occupied), or `internal`. Drives the contract-type filter.

use_class
Many2One to `real_estate.use_class`. Controls which additional fields (`basement_nr` / `parking_nr`) appear on the form.

Meter fields (visible only for `type = 'equipment'`, `e_type = 'meters'`):

meter_no_decimals
Boolean, default `True`. When set, meter readings are validated and rounded to integer values. The estimate-consumption wizard and `simulate_estimate` respect this flag and round accordingly.

Context classes for list views (`BaseObjectEquipmentContext`, `BaseObjectOccupancyContext`, `MeterReadingContext`) provide filterable search panels. Selection fields shared with the underlying model (e.g. `e_type`, `state`) are derived dynamically so their labels stay in sync automatically.

real_estate.object_party.role (object_party.py)
Configurable role definitions per object type (e.g. *Tenant* only for type `object`, *Owner* for all types).

real_estate.object_party (object_party.py)
Links a `party.party` record to a `BaseObject` with a typed role and an optional validity period.

real_estate.measurement.type (measurement.py)
Named measurement type linked to a `product.uom` unit.

Supports a **group hierarchy**: a type with `is_group = True` acts as a container grouping one or more child types via the `parent` Many2One field. Multi-level hierarchies are supported — groups can themselves have a parent group (e.g. *Gross Floor Area* → *Net Floor Area* → *Living Area*).

Constraints enforced at save time:

- All child types must share the same unit as their parent group.
- Circular references (a group referencing one of its own descendants as parent) are rejected.
- Only **leaf** (non-group) types can be assigned to individual `real_estate.measurement` records on `BaseObject` instances.

When a group type is referenced in `ContractTermType.m_type` or in `SettlementUnit.m_type`, the system automatically expands the group to all descendant leaf type IDs at query time, so measurements of any child type are included in the calculation.

real_estate.measurement (measurement.py)
Associates a numeric value and measurement type with a `BaseObject` for a given validity period.

real_estate.meter_reading (base_object.py)
Meter reading record linked to an equipment object of type `meters`.

Key fields: `company` (stored, auto-filled from `base_object` on change), `base_object`, `meter_id`, `reading_date`, `m_type` (`initial` / `reading` / `estimate` / `final`), `value`, `unit` (derived from the meter's `meter_unit`), `consumption` (difference to previous reading for counter meters), `comment` (free text).

Browseable via the *Meter Readings* menu entry under *Master Data*, filterable by company, property, parent object, equipment (meters only), and date range (`from_date` / `to_date`).

Consumption estimate (`simulate_estimate` / `create_estimate`):

simulate_estimate(base_object, per_date, meter_id=None)
Returns `(estimated_value, consumption, r1, r2)`. Prefers **interpolation** when readings exist both before and after `per_date`: takes the closest reading on each side and interpolates linearly between them. Falls back to **extrapolation** using the last two readings within

one year before `per_date` when no future reading is available. Rounds the result to the meter's UOM digit precision; if `meter_no_decimals` is set on the meter, rounds to integers.

`create_estimate(base_object, per_date, reason, meter_id=None)`
Calls `simulate_estimate` and saves a new reading with `m_type = 'estimate'`.

11.2 Contract Management

`real_estate.contract.type` (`contract_type.py`)
Template for contracts. Defines invoice direction (`in`/`out`), default taxes, accounting journal, number prefix, step sizes for item and term sequence numbers, and whether occupancy exclusivity is enforced.

`oc_mark`
Free-text label used in operating cost settlement invoice descriptions and invoice headers, e.g. `"Betriebskostenabrechnung 2025"`. Falls back to `"Operating Cost Settlement"` / `"Operating Costs"` when empty.

`real_estate.contract.type.tax` (`contract_type.py`)
Many2Many relation table between `ContractType` and `account.tax`.

`real_estate.contract.term.type` (`contract_type.py`)
Template for contract terms. Defines default rhythm, rhythm type (daily / weekly / monthly / quarterly / annually / one-time), rhythm start day, measurement type for quantity derivation, default quantity, and default account.

`real_estate.contract` (`contract_core.py`)
Main contract record.

Workflow: Draft Running Terminated / Cancelled

Key features:

- Links to `party.party` (contractual partner), `base_object` (property), and one or more `ContractItem` records
- Generates Tryton accounting moves (invoices) for all active terms via `CreateContractMoves` wizard (`call_create_moves`)
- Cash flow tabs on the contract form: *Draft* (invoice state `draft`/`validated`), *Pending* (`posted`), *Paid* (`paid`)
- `_refresh_occupancy_for_contracts` updates occupancy records when a contract starts or is terminated; cancelled contracts are excluded from occupancy calculations automatically
- `add_log` appends timestamped `ContractLog` entries
- `next_item_sequence` / `next_term_sequence` auto-increment helpers

`settlement_units` (Function field, `get_settlement_units`)
The contract's *last valid* settlement units: settlement units of the property's billing units whose objects overlap with the objects assigned to this contract via its items. Billing units are restricted to the property's `next_billing_start_date` (the earliest non-billed billing unit start date); if the property has none set, the most recently `billed` period is used instead. Used by `get_cost_shares` (cost shares of these settlement units belonging to the contract) and by `real_estate.contract.annex4.report` for the Anlage 4 print (see *Reports* below) — the report intentionally reuses this field instead of re-deriving the billing unit itself.

Fixed term / termination. `unlimited` (Boolean, default `True`) determines whether the contract has a fixed end date. Editable only in `draft`:

- `unlimited = True` (default) — `end_date` is readonly and must be empty; `on_change_unlimited` clears it automatically when toggled on.
- `unlimited = False` — `end_date` becomes required and editable (only in `draft`).
- The invariant ("unlimited contract must not have an end date", `msg_contract_unlimited_with_end_date`) is enforced by `validate_fields` only in `draft` — once `running`/`terminated` it is no longer checked, since termination legitimately sets `end_date` regardless of `unlimited` (see below).

The *Terminate* button (`real_estate.terminate_contract.wizard`, see *Wizards*) now also writes the resolved `termination_date` into `end_date` — so `end_date` always reflects the contract's actual end once terminated, fixed-term or not.

Reactivating a **terminated** contract no longer uses the *Running* button (that button is now only shown for `draft`/`cancelled`). Instead, a dedicated *Revert Termination* button (`revert_termination`, same underlying `terminated → running` transition) is shown only for `state = 'terminated'`. Besides clearing the termination fields (`termination_date`, `terminated_by_type`, `receipt_of_termination_notice`, `termination_notice`, `termination_reason` — same as *Running* already did), it additionally clears `end_date` back to empty, but **only if the contract is "unlimited"** — a fixed-term contract that was terminated early keeps whatever is in `end_date` after being reactivated.

See also the `update_contract_status` cron task (*Configuration* above) for the automatic `expired` termination of fixed-term contracts whose `end_date` has passed without an active termination.

Cancellation: the *Cancel* button transitions the contract to *Cancelled*. Before the transition:

- If any cash flow entry in state `done` (already invoiced) exists, a `ContractCancelWarning` is shown explaining that existing postings will **not** be reversed automatically. The user must confirm to proceed.
- All cash flow entries still in state `draft` are deleted.
- Occupancy records are recalculated; the cancelled contract is excluded.

`real_estate.contract.log` (`contract_core.py`)
Append-only audit log attached to a contract (event name + description).

`real_estate.contract.item` (`contract_item.py`)
Associates a `base_object` of type `object` (e.g. an apartment) with a contract for a given validity period. On create/write/delete, triggers occupancy refresh and re-runs BillingUnit selection and value-share calculation for the affected property. Validates that `valid_from` lies within the contract period and raises a warning (or error for active contracts) on overlapping occupancy.

Each `ContractItem` can reference one or more rental objects via the `real_estate.contract.item.object` child model. The object selector always restricts to objects belonging to the **same property** as the contract (preventing cross-property assignments); it is additionally restricted to `type = 'object'` only for **occupancy contracts** (`contract.c_type.occupancy` set — the Function field `occupancy` on `ContractItemObject` mirrors this from the contract type). Non-occupancy contract types may reference any `base_object` type (building, land, equipment, …) belonging to the property. A unique SQL constraint on `(item, object)` prevents assigning the same object twice to the same item; the same object may still be assigned to a different item (e.g. on another contract).

`real_estate.contract.term` (`contract_term.py`)
A recurring charge line on a contract (rent, operating cost advance, etc.).

Key fields: `term_type`, `reference_item`, `valid_from`/`valid_to`, `rhythm` + `rhythm_type` + `rhythm_start`, `quantity`, `unit`, `unit_price`, `taxes`.

Key methods:

`re_calc()`
Rebuilds the `CashFlow` list from existing invoice lines and projects future entries up to `_re_calc_year` years ahead.

`_next_document_date()`
Calculates the next invoice date based on rhythm and last posting date.

`_on_change_with_next_due_date()`
Applies the contract's payment term to derive the due date.

`real_estate.contract.term.tax` (`contract_term.py`)
Many2Many relation table between `ContractTerm` and `account.tax`.

`real_estate.contract.term.cash_flow` (`contract_term.py`)
Projected or realised cash flow entry for a term.

States: `draft` (planned, no invoice line yet) · `done` (invoiced)

Invoice states (computed from the linked invoice): `draft` / `validated` — not yet posted (excluded from settlement calculations); `posted` — open receivable; `paid` — settled. Only `posted` and `paid` entries appear in the balance sheet and are used as *advance payment* in operating cost settlement.

Stores `posting_date`, `document_date`, `due_date`, and a link to the `account.invoice.line` once billed. `base_object` (function field) is derived from the linked invoice line and used to key advance payments to a (contract, object) pair during operating cost settlement. `create_moves_run_id` (format YYYYMMDD–HHMMSS–U<userid>) is set for all cash flow entries created in a single `CreateContractMoves` run, allowing traceability back to the wizard invocation. Supports date-pattern search (`YYYY/MM` or `YYYY/MM/DD`) in the name field.

`Quantitative` (`contract_term.py`)
Custom `fields.Numeric` subclass that carries a `unit` reference and sets `quantitative=True` in its field definition so the UI renders the associated unit symbol.

`real_estate.contract.term.adjustment` (`contract_term.py`)
History record documenting a single old-term → new-term adjustment (e.g. a rent/advance-payment change following an operating cost billing run, an operating cost plan, or a free percentage/absolute change).

States: `draft` · `approved`

Key fields: `adjustment_mode` (`percentage` / `absolute`), `term_old` (required, `ondelete='RESTRICT'`), `term_new` (`ondelete='RESTRICT'`, set once the replacement term exists), and an optional `settlement_result` link when the adjustment originates from an operating cost billing run.

Function fields mirror `company` / `property` / `contract` from `term_old`, and `valid_from` / `valid_to` / `amount` / `tax_amount` / `total_amount` from both `term_old` (suffix `_old`) and `term_new` (suffix _new), so a reviewer can compare the before/after amounts without opening either term individually.

Browseable read-only via the *Contract Term Adjustments* menu entry under *Contracts* → *Adjustment*.

> **Note**
>
> Only the data model and its history/comparison fields exist so far — nothing currently creates these records. See the *Adjustment of Contract Terms* wizard (below, under *Wizards*), which defines the full input mask but still has placeholder processing logic only.

11.3 Operating Cost Settlement

real_estate.cost_category_group (billing_unit.py)
Groups cost types for reporting, e.g. *Heating*, *Water*, *Janitorial*.

real_estate.cost_type (billing_unit.py)
Individual cost item (e.g. *Gas*, *Cold Water*) with optional `category_group`, `comment`, and `no_print` flag.

For `allocation_by_consumption` settlement units, two fields control the meter-reading tolerance window around the start/end date of each cost share:

reading_pre_days
Days *before* the target date within which a reading is accepted (default: 7).

reading_post_days
Days *after* the target date within which a reading is accepted (default: 7).

The reading closest to the target date within the window is used. If no reading is found, an error is stored on the cost share.

real_estate.billing_unit (billing_unit.py)
Annual (or period) operating cost settlement for one property.

Workflow: Draft Approved Selection Value Share Ready for Billing Billed

external_billing
Boolean flag. When enabled, every settlement unit of the billing unit is forced to `allocation_rule = allocation_from_external_billing` (see `real_estate.settlement_unit` below) and costs are entered manually instead of being computed internally.

Two calculation methods:

rental_apartment
Operating cost settlement for residential tenancies under §§ 1–2 BetrKV. Costs are allocated to tenants proportionally.

WEG_billing
Annual statement for condominium owners under WEG (German condo law). Costs are allocated to co-owners by ownership share.

Key actions (buttons):

approved
Activates the billing unit.

selection
Identifies which contracts/objects are in scope for the billing period. Available in states `approved`, `selection`, `value_share`, and `ready_for_billing` (re-running from `value_share` / `ready_for_billing` revises the scope; existing settlement results are deleted after confirmation).

compute_value_shares_button
Calculates allocation shares (`CostShare`) for each settlement unit. Available in states `selection`, `value_share`, and `ready_for_billing`. If settlement results already exist for the billing unit, a confirmation warning is shown before they are deleted and value shares are recomputed. Does **not** automatically call `compute_settlement_result`. If any settlement unit has `sub_state = error` the billing unit is *not* advanced to `value_share`; the error is written to the billing unit log instead.

compute_settlement_result
Derives SettlementResult records per (contract, base_object) pair — actual costs, advances paid (only posted/paid invoice lines), and the resulting refund or additional receivable. Advance payments are matched to cost shares by the same (contract, object) key. If an advance-payment cash flow line has no matching cost-share group (e.g. no object on older invoices, or several terms for the same object) a fallback SettlementResult with actual_costs = 0 and a full refund is created for it; the fallback count is logged. Sets affected CostShare records to state error if any cash flow entries in the period are still in draft/validated state; the error message is prefixed with [draft] so it can be reset on re-run once the drafts are posted or deleted. Resets a billing unit already in state ready_for_billing back to value_share before recomputing.

check_ready_for_billing
Transitions value_share → ready_for_billing after validating:

- no settlement unit has sub_state = error;
- every settlement unit (except no_allocation) has reached sub_state = value_share;
- at least one approved SettlementResult exists and none has actual_costs = 0;
- every settlement result with a non-zero advance payment has both a term and a contract (a missing term usually means several term types contributed and the billing unit's term-type filter needs narrowing);
- all advance-payment invoices are posted;
- (unless external_billing) the sum of settlement result actual costs matches the sum of cost share actual costs.

Any failed check raises a ValidationError listing the offending records. Can also be triggered per property via BaseObject.ready_for_billing_property for all billing units sharing next_billing_start_date.

billing_wizard / billing
The form button billing_wizard opens the real_estate.billing_unit.wizard (see *Wizards* below), which calls the billing classmethod. billing creates, per settlement result, up to two invoice lines — advance-payment dissolution and actual costs — with taxes copied from the advance-payment term, then one invoice per contract. Invoices are created in state draft and posted immediately if the wizard's invoice_state is posted. Each invoice's payment_term (and therefore its due date) is taken from the wizard's payment_term field when set; otherwise the contract's own payment term is used, falling back to account.configuration's re_payment_term_billing. Both invoice_date and accounting_date are set to the wizard's invoice_date (analogous to the accounting_date on contract invoices generated by CreateContractMovesWizard), so the posting date matches the invoice date rather than the date the wizard was run. Refuses to proceed if sub_state is error or if draft cash flow lines still exist in the settlement period; with collective_billing all billing units of the property sharing the same start date must already be ready_for_billing. Generates a billing_run_id (YYYYMMDD-HHMMSS-U<userid>) that is written to the billing unit and all BillingUnitMoves records of the same run, so every posting can be traced back to its run. Hidden when the property has collective_billing = True; in that case the action is triggered from the property form instead (BaseObject.billing_property → same wizard → BaseObject.do_billing).

billing_run_id
Char field set at the end of a successful billing() call, identical to the value written to all BillingUnitMoves of that run.

real_estate.billing_unit.log (billing_unit.py)
Timestamped log entries attached to a billing unit.

real_estate.billing_unit.moves (billing_unit.py)
One record per invoice line created by `billing()` — a separate record for the advance-payment dissolution line and for the actual-costs line of the same settlement result (not combined into one record). Links the billing unit to the contract, settlement result, and the individual posting record:

- `moves_advanced_payment` — `account.invoice.line` that reverses the operating-cost advance (credit note line)
- `moves_actual_costs` — `account.invoice.line` for the actual costs
- `moves_alloc_by_owner` — `account.move.line` for the owner allocation journal entry

Function fields `amount`, `currency`, and `invoice` are derived from whichever of `moves_advanced_payment` / `moves_actual_costs` is set, for quick display in the tree view without opening the invoice line. For vacancy records (no `contract`, but `moves_alloc_by_owner` set), `amount` is instead computed as `debit - credit` of that `account.move.line`, `currency` from its own `currency` field, and `invoice` stays empty (move lines have no invoice reference).

`billing_run_id` ties all moves of one billing run together (same value as on the parent `BillingUnit`). Browseable via the *Billing Unit Moves* menu entry under *Operation Costs*, filtered by company, property, billing unit, contract, and date range.

real_estate.settlement_unit (settlement_unit.py)
One cost-type line within a billing unit, e.g. *Gas 2024*.

Allocation rules:

no_allocation
Entire cost stays unallocated.

allocation_by_measurement
Allocated by a measurement value, e.g. living area. `value_share` = `measurement_value × time_share / time_total` (time-weighted: a tenant occupying only part of the period receives a proportionally smaller share).

allocation_by_consumption
Allocated by meter reading consumption (HeizkostenV). `value_share` = raw consumption (no time weighting applied).

Meter readings are looked up within a tolerance window defined by `reading_pre_days` / `reading_post_days` on the cost type; the closest reading to the target date is chosen.

Vacancy handling: cost shares without a contract receive `value_share = 0` automatically — no meter reading is required. For the immediately following contract cost share, if no start reading exists within the normal window, the system falls back to the reading taken at the start of the preceding vacancy. This allows a single read-out at move-out to serve as both the vacancy baseline and the contract's opening reading.

Missing readings set the cost share to state `error` with one of:

- *"Messwert für Anfangsverbrauch nicht ermittelt"* — start reading absent
- *"Messwert für Endverbrauch nicht ermittelt"* — end reading absent

allocation_per_rental_unit
Proportional share by occupancy duration. `value_share` = `time_share / time_total` (full period = 1.0, half period = 0.5, etc.).

allocation_from_external_billing
No internal calculation; `planned_costs`/`actual_costs` are entered manually on the cost

share. Automatically set on every settlement unit of a billing unit when that billing unit's `external_billing` flag is enabled (see `real_estate.billing_unit` below); the settlement unit's `allocation_rule` then becomes read-only.

Vacancy handling: unoccupied periods can be charged to the owner or left unallocated.

When `compute_value_shares` is re-run, cost shares in state `error` are automatically reset to `selection` and their `error_message` cleared before the new calculation starts, so corrected readings or measurements take effect immediately.

`real_estate.cost_share` (`settlement_result.py`)
Intermediate allocation record: share of a specific cost type for one contract or object within a billing period.

States: `preparation` · `selection` · `estimated_value_share` · `value_share` · `error`

Stores `value_share` (time-weighted allocation factor — for `allocation_by_measurement` and `allocation_per_rental_unit` this already incorporates the occupancy fraction `time_share / time_total`; for `allocation_by_consumption` it holds the raw consumption value), `time_share` (days), `planned_costs`, `actual_costs`. `error_message` describes the problem; entries set by `compute_settlement_result` carry a `[draft]` prefix and are automatically cleared on the next successful re-run.

`allocation_rule` / `external_billing` (Function fields, mirrored)
Mirrored from the parent `settlement_unit`. `planned_costs`/`actual_costs` are only editable (`readonly` otherwise) when `external_billing` is set, i.e. for cost shares of a settlement unit with `allocation_rule = allocation_from_external_billing`.

`real_estate.settlement_result` (`settlement_result.py`)
Final settlement record per contract (or object) and billing unit.

States: `approved` · `billed`

Stores `planned_costs`, `actual_costs`, `advanced_payment` (sum of `posted`/`paid` operating-cost advance invoice lines during the period), and the derived `refund_receivable` (positive = refund to tenant, negative = additional receivable). Optionally links to the single `ContractTerm` that covered all advances when exactly one term was involved. The display name includes the record id in parentheses (e.g. `"2026-01-01 – 2026-12-31 / Mieter 1 (42)"`) to disambiguate several results for the same contract/object; the id is also searchable via the name field.

11.4 CO2 Cost Allocation (CO2KostAufG)

Implements the German *CO2-Kostenaufteilungsgesetz* (CO2KostAufG), which splits the CO2 cost of heating fuel between tenant and landlord depending on the building's emission intensity (kg CO2/m²/year). Applies to `real_estate.settlement_unit` (typically the *Heizkosten* settlement unit of a billing unit) via an optional reference to a `real_estate.co2_kostaufg` master record — settlement units without a reference are unaffected.

`real_estate.co2_kostaufg` (`co2_kostaufg.py`)
Master record for one CO2KostAufG data source (e.g. one heating installation/fuel), scoped to a `property` (required, `type = 'property'`) with `company` derived from it (Function field). Referenced from one or more settlement units of the same property via their `co2_kostaufg` field (domain-restricted to that property — a settlement unit can only pick a CO2KostAufG belonging to its own property). Menu: *Real Estate* → *Operation Costs* → *CO2 Kosten verwalten*.

`real_estate.co2_kostaufg.consumption` (`co2_kostaufg.py`)
Child records under a `co2_kostaufg` (`parent`, `ondelete='CASCADE'`), one per billed/metered

period: `date_from`/`date_to`, `consumption_kwh`, `co2_kg_per_kwh` (emission factor), `co2_emission_kg`, `co2_price_ct_per_kwh`, `vat_rate`, `co2_cost_net`, `co2_cost_gross`.

`co2_emission_kg` is pre-filled (`consumption_kwh` × `co2_kg_per_kwh`) only while still empty — once any value exists (auto-filled or typed by the user), later changes to `consumption_kwh`/`co2_kg_per_kwh` never overwrite it again, so it is always freely (re-)editable, e.g. to enter the supplier's own figure. `co2_cost_net`/`co2_cost_gross` behave differently **on purpose**: they are always recomputed from `consumption_kwh`/`co2_price_ct_per_kwh`/`vat_rate` whenever any of those three change; a manual override of net/gross only sticks until the next such recompute.

`split`
Boolean. Set automatically by `SettlementUnit.compute_value_shares()` when this record was created by splitting an original record at a settlement-period boundary — see *Splitting at settlement-period boundaries* below.

Uses `DeactivableMixin` — a record replaced by a split is deactivated (`active = False`), never physically deleted.

`real_estate.co2_emission_share` (`co2_kostaufg.py`)
Configuration table for the legal 10-tier distribution model (Anlage 1 CO2KostAufG, residential rented buildings without separate sub-metering by usage type): `emission_limit` (kg CO2/m²/year — the *exclusive* upper bound of the tier), `tenant_share` / `landlord_share` (%, must sum to 100 — enforced in `validate()`). `get_share(value)` returns the first row, in `sequence` order, whose `emission_limit` is strictly greater than `value`; if `value` reaches or exceeds every configured limit, the last row (by sequence) is used as an open-ended top tier regardless of its own stored limit value. Default data for all 10 legal tiers is loaded at module installation (< 12 kg 0 % / 100 % landlord/tenant … ≥ 52 kg 95 % / 5 %). Menu: *Real Estate* *Configuration* *Emission-CO2 Verteilung*.

Fields added to `real_estate.settlement_unit` — a new *CO2 Costs* page is always shown, but every field on it except `co2_kostaufg` itself stays hidden (`invisible`) until a reference is actually set:

`co2_kostaufg`
Many2One reference, restricted to CO2KostAufG records of the same `property` as the settlement unit.

`co2_measurement_type`
Many2One to `real_estate.measurement.type` (object-type measurements only) — which measurement to sum across the settlement unit's `objects` for the area basis below. Independent of the settlement unit's own `m_type` (used for `allocation_by_measurement`).

`co2_total_consumption` / `co2_total_emission` / `co2_total_cost_gross`
(Function.) Sum of `consumption_kwh` / `co2_emission_kg` / `co2_cost_gross` across all consumption records of the referenced CO2KostAufG that overlap the settlement period (the parent billing unit's `start_date`/`end_date`), each weighted by the fraction of the record's *own* date range that actually falls inside the period. A record fully inside the period contributes 100 %; a record extending beyond either boundary is interpolated pro-rata by days, always relative to its own actual range — a record that simply doesn't reach as far as the period's end (no later data booked yet) is never extrapolated, it only ever contributes its own days.

`co2_total_area`
(Function.) Sum, across the settlement unit's `objects`, of each object's latest `real_estate.measurement` value of `co2_measurement_type` (or its effective leaf types, if a measurement group) valid as of `end_date`.

co2_emission_per_m2
(Function.) `co2_total_emission / co2_total_area`, annualized via `× 365 / time_total` so that settlement periods shorter or longer than a calendar year still yield a correct "kg CO2/m²/year" figure.

co2_tenant_share / co2_landlord_share
(Function.) Looked up via `Co2EmissionShare.get_share(co2_emission_per_m2)` — the residential 10-tier split.

co2_commercial_tenant_share / co2_commercial_landlord_share
(Function.) The flat split used for commercial properties (not covered by the residential tier model): the company's configured `re_accounting.co2_landlord_share_commercial` (landlord), and `100 %` minus that value (tenant).

co2_consumption
(Function, One2Many, readonly.) All consumption records of the referenced CO2KostAufG that overlap the settlement period — shown as an embedded, read-only table at the bottom of the *CO2 Costs* page.

Splitting at settlement-period boundaries. `SettlementUnit.compute_value_shares()` ("Compute Value Shares" button, also reachable via the billing unit's own button of the same name) calls `_split_co2_consumption()` first, before anything else. For each of `start_date` and `end_date + 1 day`, every active consumption record of the referenced CO2KostAufG whose own `[date_from, date_to]` strictly contains that boundary date is split into two new records at that date: amounts are interpolated pro-rata by day count (the second part is computed as the remainder of the first, so the two new records always sum back exactly to the original — no rounding drift); rate fields (`co2_kg_per_kwh`, `co2_price_ct_per_kwh`, `vat_rate`) are copied unchanged onto both. Both new records get `split = True`; the original is deactivated (soft-deleted via `active`), never physically removed. A record entirely inside the settlement period, or one that only covers part of it because no later data has been booked yet, is never split or extrapolated — only an *existing* record that actually spans across a period boundary gets divided. Re-running is idempotent: already boundary-aligned active records are left untouched, so repeated calls never produce duplicate splits.

11.5 Option Rate (Input VAT Deduction)

Tracks, over time, what percentage of input VAT (Vorsteuer) on purchase invoices related to a real estate object is deductible ("optiert") rather than treated as a final cost (added to the gross expense) — relevant wherever the property owner can opt for VAT liability on rent (§ 9 UStG). Applies to `real_estate.base_object` (property / building / land / rental object), `real_estate.settlement_unit`, and `real_estate.billing_unit`.

account.invoice.line.taxes_deductible_rate auto-fill (invoice.py)
For purchase invoice lines (`invoice_type = 'in'`), the core `taxes_deductible_rate` field (0–1 fraction of input VAT that is deductible; core already forces it to `0` when `company.purchase_taxes_expense` is set, via its own `on_change_company`) is auto-filled from the applicable option rate, divided by 100. Applies both interactively (`on_change_base_object` / `on_change_settlement_unit` / `on_change_billing_unit`) and to programmatic line creation (`InvoiceLine.create()` override) — the latter only fills the field when the caller did **not** pass `taxes_deductible_rate` explicitly, so an explicit value is never overridden.

The real-estate reference used is picked by `InvoiceLine._option_rate_priority()` in order of specificity: `base_object` → `settlement_unit` → `billing_unit` → the derived `property` (as a `base_object` reference, last-resort fallback via `on_change_with_property` / `term.property`) — the first of these that is set on the line wins. `OptionRate.get_current_rate_fraction(ref_field, record, date)` then returns the record's latest `option_rate` at or before that date (`taxes_date` if set, else the invoice's `invoice_date`, else

today), divided by 100; falling back to 1/0 for fix_100/fix_0 if no history record exists yet, or None (leaving the field untouched) for dynamic_measurement with no booked history.

The auto-fill is skipped entirely (field left at its core default of 1) when: the line is not a purchase line, or the company has purchase_taxes_expense set (core's own logic already governs that case). The line's contract field has no bearing on this logic and is ignored — a real-estate reference is used whenever present, even on a line that also carries a contract.

real_estate.option_rate (option_rate.py)
History-versioned option rate record, valid from a given date. Has three mutually exclusive Many2One reference fields — base_object, settlement_unit, billing_unit (all ondelete='CASCADE') — exactly one of which must be set (enforced in validate()); a unique SQL constraint prevents two records for the same reference and valid_from. option_rate is a percentage (digits=(5, 2), domain 0–100).

Fields added to real_estate.base_object, real_estate.settlement_unit, and real_estate.billing_unit (identical on all three):

option_rate_method
Required selection: fix_0 (always 0 %), fix_100 (always 100 %), or dynamic_measurement (calculated — see the wizard below). Defaults to fix_0. On a rental object (base_object with type = 'object'), changing type_of_use autosets it to fix_100 for commercial and fix_0 otherwise (on_change_type_of_use); dynamic_measurement is rejected by validate_fields for rental objects — only property/building/land/ settlement unit/billing unit may use it.

option_measurement_type
Many2One to real_estate.measurement.type, required only for dynamic_measurement. May reference a measurement **group**: all of its descendant leaf types are then summed per rental object (see the group hierarchy under real_estate.measurement.type above).

option_rates
One2Many (readonly) to the object's own real_estate.option_rate history.

purchase_taxes_expense
Function field mirroring the pre-existing company.purchase_taxes_expense field (added by account_invoice). When set on the object's company, all input VAT is always booked as an expense (fully gross) — equivalent to a permanent option rate of 0 % — and the *Option Rate* notebook page is hidden on the form (also always hidden for type = 'equipment').

Shown as a dedicated *Option Rate* page on the base_object, settlement_unit, and billing_unit forms (page_option_rate).

real_estate.option_rate_update.wizard (see *Wizards* below)
Batch recalculation entry point; see the *Wizards* section for the full selection/calculation logic.

Selection / filter screen and standalone list. Menu: *Real Estate* → *Master Data* → *Update Option Rates*, with a child menu item *Option Rates* opening a read-only list of all real_estate.option_rate records (both list and form views are fully non-editable — creatable="0" plus readonly="1" on every field; this applies only to the standalone list, not to the option_rates tab embedded in the base_object/settlement_unit/billing_unit forms, which remains editable there). A *Selektion* context form — modelled on the *Occupancy* context pattern (real_estate.option_rate.context, class OptionRateContext) — sits above the list with three optional filter fields, base_object, billing_unit, settlement_unit; whichever one is set narrows the list via context_domain (PYSON, evaluated independently per field, so any combination — or none — can be applied).

11.6 Extensions to Core Modules

party.party (party.py)
Adds sequence (integer sort key) and salutation fields.

account.invoice (invoice.py)
Adds a contract Many2One field so invoices can be traced back to the originating contract. Shown on the invoice form's *Other Info* tab (view extension view/invoice_form.xml, inherits account_invoice.invoice_view_form).

account.invoice.line (invoice.py)
Adds real-estate assignment fields, gated by assignment_control (Selection: *All*, contract, operating_costs, settlement_result_contract, settlement_result_vacant — controls which of the fields below are visible/required for a given line):

- contract / term — the originating real_estate.contract / ContractTerm
- base_object — the real-estate object the line relates to
- billing_unit / settlement_unit — for operating-cost billing lines
- property (Function) — derived from billing_unit/settlement_unit
- service_period_from / service_period_to — billed service period
- estg_35a (Selection) — German §35a EStG tax-deduction category (household services / craftsmen services)
- invoice_date, tax_amount, total_amount (Function fields)

Indexed on contract, term, settlement_unit, billing_unit.

account.move.line (invoice.py, class AccountMoveLine)
Mirrors the same real-estate fields as account.invoice.line above (assignment_control, contract, term, base_object, billing_unit, settlement_unit, property) directly on the journal entry line, so postings can be traced/filtered without going through the invoice line. Extended list view shows this context for journal entries.

account.general_ledger.line (invoice.py, class GeneralLedgerLine)
Adds contract, term, base_object, billing_unit, and settlement_unit (all readonly Many2One) to Tryton's standard General Ledger — Lines report (German: *Kontenblätter – Positionen*), so the individual postings for an account can be filtered/traced back to the originating real-estate contract, term, object, billing unit, or settlement unit. No override of table_query is needed: it pulls any non-Function field straight from the account.move.line table by matching field name, and account.move.line already carries these same fields (see above). Shown as optional columns in the tree view (view/general_ledger_line_list.xml, inherits account.general_ledger_line_view_list).

account.configuration (account_configuration.py)
Extends the standard account configuration with real-estate-specific defaults used during operating cost billing:

re_account_allocation_by_owner
Vacancy cost account (debit and credit side of vacancy postings).

re_journal_billing
Journal used for direct GL postings in vacancy settlements.

re_payment_term_billing
Default payment term for operating cost settlement invoices. Pre-fills the payment_term

field of the `real_estate.billing_unit.wizard`; also used directly by `BillingUnit.billing()` as a final fallback when neither the wizard's `payment_term` nor the contract's own payment term is set.

All three are per-company `MultiValue` fields backed by `account.configuration.real_estate`.

`res.user` (`res.py`)
Adds `phone` and `mobile` fields.

CHAPTER

TWELVE

WIZARDS

real_estate.contract.create_moves.wizard (contract_wizard.py, class CreateContractMovesWizard)

Batch invoice generation wizard. Supports three actions:

create
: Generate invoices up to a given date. All `ContractTermCashFlow` entries created in this run share the same `create_moves_run_id` (`YYYYMMDD-HHMMSS-U<userid>`).

re_calc
: Rebuild cash flow projections without creating invoices.

re_calc_and_create
: Combine both steps.

Can be filtered by property and/or contract list. Optional queue execution via `execute_in_queue`.

real_estate.billing_unit.wizard (billing_unit_wizard.py, class BillingUnitWizard)

Batch billing wizard, opened via the `billing_wizard` button on the billing unit form or the `billing_property` button on the property form.

Start — `date` (default: last day of the current month), `company`, `invoice_date` (default: today; auto-set to the first of `date`'s month when `date` lies in the past), `invoice_state` (`draft` / `posted` — invoices are posted immediately when `posted`), `payment_term` (default: `account.configuration`'s `re_payment_term_billing`; written to every invoice created by the run and therefore drives its due date — takes priority over the contract's own payment term), `execute_in_queue` (default `True`), optional `propertys` / `billing_units` filters (pre-filled from the active record when launched from a property or billing unit form/list).

Confirm — read-only summary of the matching billing unit count and the selected filters before processing.

Process (`transition_do_billing`) — calls `BaseObject.call_billing` for the resolved properties, which invokes `BaseObject.do_billing` either directly or via the background queue. Only billing units already in state `ready_for_billing` are billed; with `collective_billing` all billing units of a property sharing the same start date must be included, otherwise a `ValidationError` is raised.

Result — reports how many billing units were queued or processed.

real_estate.terminate_contract.wizard (contract_wizard.py)

Sets a contract to *Terminated*, records `terminated_by`, `receipt_of_termination_notice`, `termination_reason`, and calculates `termination_date` from the notice period (3 / 6 / 9 / 12 months to end of month); also writes the resolved `termination_date` into the contract's `end_date` (see *Fixed term / termination* under `real_estate.contract` above). `terminated_by_type` is `tenant` or `landlord` for a manual termination via this wizard; see the `update_contract_status`

cron task above for the automatic `expired` case (fixed-term contracts with no active termination), and the *Revert Termination* button to undo a termination.

`real_estate.estimate_consumption.wizard` (`base_object.py`, class `EstimateConsumptionWizard`)

Two-step wizard launched from the *Meters* tab of any approved meter object. Estimates the meter reading for a chosen cut-off date and books it as an `estimate` reading.

Step 1 – Start: displays the meter description, unit and factor, pre-fills `meter_id` from the last reading, and prompts for `per_date` (default: today) and a free-text `reason`.

Step 2 – Result: calls `simulate_estimate` and shows the two basis readings (`reading1` / `reading2`), the derived `consumption`, and the editable `estimated_value`. The user can adjust the value before booking.

Book: saves a new `MeterReading` with `m_type = 'estimate'`. The value is rounded to the meter's UOM digit precision (integer if `meter_no_decimals` is set) before saving.

`real_estate.option_rate_update.wizard` (`option_rate_wizard.py`, class `OptionRateUpdateWizard`)

Recomputes and books option rates (see *Option Rate* above) for a selection of base objects, billing units, and/or settlement units, as of a cut-off date.

Start — `base_objects` (Many2Many, restricted to property/building/land/rental object; an approved property/building/land is expanded to its building/land/rental-object descendants, and an approved property additionally cascades to its billing units and their settlement units), `billing_units` (expanded to their settlement units), `settlement_units`, `cutoff_date` (default: today — booked option rates are always dated on the first of this month).

Confirm — read-only counts of the directly selected base objects / billing units / settlement units, and the total number of objects that will actually be processed after expansion (`n_expanded`).

Process (`OptionRate.process_update`) — for every expanded object:

- Objects with `option_rate_method` `fix_0` / `fix_100` are booked at 0 % / 100 % directly, without looking at any sub-objects.
- `dynamic_measurement` objects are calculated from their approved rental objects: for a rental object, itself; for a property/building/ land, its approved rental-object descendants regardless of how many buildings/land lie in between (a non-approved intermediate object excludes its whole subtree — both from booking and as a calculation basis); for a settlement unit, its existing `objects` function field (i.e. whatever it already resolves to via its own allocation rule/regex — read-only here, never modified by this wizard); for a billing unit, the union of `objects` across all its settlement units that are not `draft`/`billed`, deduplicated. Each contributing rental object is weighted by its measurement value (leaf types under `option_measurement_type` summed if it is a group) at the cut-off date, and contributes its own current option rate (its latest `option_rate` record at or before the cut-off date, else its own `fix_0`/`fix_100` default). The new rate is the resulting weighted average, rounded to 2 decimals.
- Base objects not in state `approved`, and billing/settlement units in state `draft` or `billed`, are excluded entirely.
- A new dated `option_rate` record is only written when the computed rate differs from the current one; a record already dated on the cut-off month's first day is updated in place rather than duplicated.

Result — counts of created / updated / unchanged / skipped records, plus a per-record detail log.

Menu: *Real Estate* → *Master Data* → *Update Option Rates*.

`real_estate.contract_term_adjustment.wizard` (`contract_wizard.py`, class `ContractTermAdjustmentWizard`)

Note

Skeleton only. The *Start* → *Confirm* → *Process* → *Result* flow and every field below already work end-to-end, but the actual selection of contracts/terms and the write-back to `ContractTerm` / `real_estate.contract.term.adjustment` are not implemented yet. `transition_do_adjustment` dispatches by `procedure` to one of three placeholder methods (`_adjustment_operation_costs_billing`, `_adjustment_operation_costs_plan`, `_adjustment_free_adjustment`), each of which currently just returns `processed = 0` and a "not yet implemented" message without changing any data.

Start — `procedure` (`operation_costs_billing` / `operation_costs_plan` / `free_adjustment`, required), `company`, `property` (defaulted from the active property record when launched from a property form/list), `valid_from_new` (effective date of the replacement term — the old term is intended to close the day before). Only for `free_adjustment`: `adjustment_mode` (`percentage` / `absolute`). Only for `operation_costs_billing`: `billing_run_id` (Selection, populated from `billed` billing units matching `company`/`property`), guard flags `no_terminated_contracts`, `no_future_terms`, `no_booked_terms`, `only_rhythm_monthly_1` (all default `True`), and caps `max_adjustment_percent` (default 10 %) / `max_adjustment_absolute` (default 50, in the company currency).

Confirm — read-only echo of every *Start* value (no match count, since selection logic does not exist yet).

Result — `processed` count and a details message.

Menu: *Real Estate* → *Contracts* → *Adjustment* (submenu with the wizard itself and a read-only list of `real_estate.contract.term.adjustment` records).

CHAPTER

THIRTEEN

REPORTS

ODT templates (OpenDocument Text) are located in `report/` and rendered via Genshi/relatorio; `template_extension` on the `ir.action.report` record is `odt`. Values are inserted with `text:span py:content="..."` rather than literal `${...}` interpolation (which relatorio escapes in ODT text), and control flow (`py:if`/`py:for`/`py:choose`) is written as attributes on the surrounding ODF elements. Superseded `.html` versions of these templates remain in `report/` for reference but are no longer registered.

`real_estate.contract.report` (`contract_report.py`)
: General contract report. Templates: `contract_en.odt`, `contract_letter_de.odt`.

`real_estate.contract.annex4.report` (`contract_report.py`)
: Annex 4 – Betriebskostenaufstellung. Template: `anlage4_contract_de.odt`. Renders grouped operating cost positions with BetrKV paragraph references and allocation method labels, sourced from the contract's own `settlement_units` field (see `real_estate.contract` below) rather than re-deriving the billing unit independently. Allocation labels are translatable messages (`real_estate.msg_allocation_*` in `message.xml`).

`real_estate.base_object.report` (`base_object.py`)
: Fact sheet for a property or object. Template: `fact_sheet.odt`.

CHAPTER

FOURTEEN

ACCOUNTING / WOWI

The `wowi/` directory contains a German “Kontenrahmen der Wohnungswirtschaft” (WoWi, housing-industry chart of accounts) as XML templates loaded at module installation:

- Account types and accounts
- Tax groups, tax templates, tax code templates, tax code line templates
- Tax rule templates

https://www.intex-publishing.de/pdf/Kontenrahmen_Wohnungswirtschaft.pdf https://www.intex-publishing.de/pdf/Kontenrahmen_Immobilienwirtschaft.pdf

CHAPTER

FIFTEEN

SOURCE LAYOUT

```
real_estate/
├── address.py                # real_estate.address
├── base_object.py            # real_estate.base_object, occupancy, meter␣
↪readings
├── billing_unit.py           # real_estate.billing_unit, billing_unit.moves,
│                             #   billing_unit.log, cost_type,␣
↪cost_category_group
├── billing_unit_wizard.py    # real_estate.billing_unit.wizard (batch billing)
├── co2_kostaufg.py           # real_estate.co2_kostaufg(.consumption),␣
↪co2_emission_share
├── company.py                # extension to company.company (re_accounting␣
↪link)
├── contract_core.py          # real_estate.contract, contract.log, account␣
↪views,
│                             #   cron_daily dispatcher + cron task handlers
├── contract_item.py          # real_estate.contract.item, contract.item.object
├── contract_term.py          # real_estate.contract.term, cash_flow,␣
↪Quantitative
├── contract_type.py          # real_estate.contract.type, term.type
├── contract_report.py        # ContractReport, ContractAnnex4Report
├── contract_wizard.py        # wizards (CreateContractMovesWizard,␣
↪TerminateContractWizard, …)
├── cron_task.py              # real_estate.cron_task (per-company scheduled␣
↪task config)
├── invoice.py                # extensions to account.invoice / invoice.line
├── ir.py                     # extension to ir.cron (registers the cron_daily␣
↪method)
├── measurement.py            # real_estate.measurement.type, measurement
├── object_party.py           # real_estate.object_party, object_party.role
├── option_rate.py            # real_estate.option_rate, option_rate.context
├── option_rate_wizard.py     # real_estate.option_rate_update.wizard
├── party.py                  # extension to party.party
├── re_accounting.py          # real_estate.re_accounting (company-scoped RE␣
↪config)
├── res.py                    # extension to res.user
├── settlement_result.py      # real_estate.settlement_result, cost_share
├── settlement_unit.py        # real_estate.settlement_unit
```

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```
├── report/                  # ODT report templates (rendered via␣
↪Genshi/relatorio)
├── view/                    # XML form and tree view definitions
├── wowi/                    # German WoWi accounting templates
└── locale/                  # Translations (de.po)
```

CHAPTER

SIXTEEN

RUNNING TESTS

> **Warning**
>
> `tests/` currently has no `unittest.TestCase`-based tests — `test_module.py` (the standard Tryton `ModuleTestCase` boilerplate) has been removed and not replaced. `tox.ini` still defines a real test matrix (`envlist = {py39,py310,py311,py312,py313}-{sqlite,postgresql}`) and the commands below run without error, but `unittest/xmlrunner` discovery currently finds **zero** test cases in every environment and exits **0 (success)** regardless — i.e. `tox` reports a false-positive pass, not "no tests configured". There is no CI pipeline in this repo currently invoking it automatically, but running it manually and reading "OK" is misleading until a real test case is added back.

```
# Full test matrix (sqlite + postgresql, py39–py313) — currently a
# false-positive pass, see warning above
tox

# Single environment
tox -e py311-sqlite

# Direct run
export TRYTOND_DATABASE_URI=sqlite://
export DB_NAME=:memory:
coverage run --omit=*/tests/* -m xmlrunner discover -s tests
coverage report
```

16.1 Demo data scripts

Everything in `tests/` is instead a one-shot **proteus import script** that generates demo data against a running trytond server. These are run manually — `python tests/<script>.py --database <db> [--config trytond.conf]` — not via tox/xmlrunner, in the following order (each step builds on the previous one's data):

`tests/test_immo.py`
: No prerequisites. Creates two properties (*Musterstraße 1-4* and *Musterstraße 5-8*), each with four buildings. Every building has one ground-floor retail unit (`type_of_use='commercial'`, 140 m²) and four apartments (`type_of_use='residential'`, 57/83 m² alternating) — 16 apartments, 4 retail units, and 20 water meters per property in total — plus one land parcel with four parking spaces. Meter names follow the pattern `Wasser Zähler NN` / `Wasser Zähler EHNN`, each with an initial and a consumption reading. Looks up `real_estate.use_class` by `sequence` (language-independent) and measurement types by German name — the `admin` user must be set to German:

```
python tests/test_immo.py --database <db> [--config trytond.conf]
```

tests/test_contracts.py

Requires `test_immo.py`. Creates residential lease contracts for 15 of 16 apartments per property (one left vacant), each with rent, operating cost, and heating terms; three contracts per property are terminated, two of them with a follow-up contract. The four parking spaces per property are split between existing apartment contracts (as an extra contract item) and new standalone parking contracts. Also creates commercial lease contracts for the retail units — one combined contract for all four retail units on *Musterstraße 1-4*, one contract per unit on *Musterstraße 5-8*:

```
python tests/test_contracts.py --database <db> [--config trytond.conf]
```

tests/test_billing_unit.py

Requires `test_immo.py`. Creates two billing units per property: *Kalte Betriebskosten* (measurement- and consumption-based settlement units, vacancy allocated to the owner) and *Heizkosten* (`external_billing=True`, settlement units use `allocation_from_external_billing`). Billing units are left in state `draft` — anything that requires a later state (e.g. `test_invoices.py`'s settlement-unit lookup) needs the workflow advanced manually first:

```
python tests/test_billing_unit.py --database <db> [--config trytond.conf]
```

tests/test_kreditor.py

No prerequisites. Creates 11 supplier/creditor parties (property tax office, building insurer, utilities, cleaning, caretaker, gardening, chimney sweep, heating maintenance), each with an invoice/delivery address:

```
python tests/test_kreditor.py --database <db> [--config trytond.conf]
```

tests/test_invoices.py

Requires `test_kreditor.py`, `test_immo.py`, and `test_billing_unit.py` (with billing units advanced past `draft` so settlement units can be looked up). Creates the corresponding purchase invoices — one-off and recurring — for each property, linking each line to its `settlement_unit` where a match is found. Invoices are only saved, not posted (remain in state `draft`):

```
python tests/test_invoices.py --database <db> [--config trytond.conf]
```

tests/test_payment.py

Requires `test_contracts.py` and at least one run of the `CreateContractMoves` wizard so posted tenant invoices exist. Books one payment receipt per tenant/commercial-tenant party (debit account 1800 Bank / credit the receivable account taken from that party's open lines) and reconciles the open items:

```
python tests/test_payment.py --database <db> [--config trytond.conf]
```